

Block 2: Progressions, Matrices and Determinants

Short explanations, worked examples and self-check questions for every unit, with every idea also explained in Malayalam.

ഓരോ ആശയവും ലളിതമായ ഇംഗ്ലീഷിലും മലയാളത്തിലും, ഉദാഹരണങ്ങളോടെ.

UNIT 5

Progressions

ഈ unit-ൽ sequence എന്താണെന്നും, അതിലെ രണ്ട് പ്രധാന തരങ്ങളായ Arithmetic Progression (A.P.), Geometric Progression (G.P.) എന്നിവയുടെ nth term, sum എന്നിവ എങ്ങനെ കണ്ടുപിടിക്കാമെന്നും പഠിക്കുന്നു. അവസാനം Σ (sigma) notation-ഉം ചില special sums-ഉം നോക്കുന്നു.

Sequence as a function

Ways of representing a sequence

Arithmetic Progression (A.P.)

General term of an A.P.

Sum of n terms of an A.P.

Geometric Progression (G.P.) and its general term

Sum of n terms of a G.P.

Sum of an infinite G.P.

Concept of summation (Σ) and change of origin

Sums of some special sequences

Sequence as a function

Sequence: A function whose domain is the set of natural numbers $\mathbb{N}$; its value $a_n = f(n)$ is called the nth term.

Domain ആയി natural numbers-ന്റെ set $\mathbb{N}$ ഉള്ള ഒരു function ആണ് sequence; n-ലെ value $a_n = f(n)$ -നെ nth term എന്ന് പറയുന്നു.

Real sequence: A sequence whose range is a subset of $\mathbb{R}$.

Range $\mathbb{R}$ -ന്റെ ഒരു subset ആയ sequence ആണ് real sequence.

In everyday language a sequence is things arranged in a definite order, like the multiples of 5: 5, 10, 15, 20, ... Mathematically, a **sequence** is a function whose domain is the set of natural numbers $\mathbb{N}$ and whose range may be any set. The value at n is written a_n and is called the n th term. If the range is a subset of $\mathbb{R}$, the sequence is a **real sequence**; this unit deals only with real sequences.

മലയാളത്തിൽ

Sequence എന്നാൽ ഒരു നിശ്ചിത ക്രമത്തിൽ നിരത്തിയ സംഖ്യകൾ. പക്ഷേ ഗണിതത്തിൽ ഇത് ഒരു function ആണ്: input ആയി 1, 2, 3, ... എന്ന natural numbers മാത്രം കൊടുക്കുന്നു, output ആണ് $a_1, a_2, a_3, \dots$ എന്ന terms. ഒരു ക്യൂവിൽ നിൽക്കുന്നവരെ പോലെ ആലോചിക്കുക: 1-ാം സ്ഥാനം, 2-ാം സ്ഥാനം... ഓരോ സ്ഥാനത്തിനും ഒരാൾ. സാധാരണ real function-ന് domain $\mathbb{R}$ -ന്റെ ഏത് subset-ഉം ആകാം, sequence-ന് domain എപ്പോഴും $\mathbb{N}$ ആണ് എന്നതാണ് വ്യത്യാസം.

Sequence: $f: \mathbb{N} \rightarrow S$, with $a_n = f(n)$

Notation: $a_1, a_2, a_3, \dots, a_n, \dots$ or $\{a_n\}$ or $\langle a_n \rangle$

a_n = n th term (general term)

EXAMPLE

The function $f: \mathbb{N} \rightarrow \mathbb{R}$, $f(n) = 1/n$. Write the sequence it represents.

$f(1) = 1, f(2) = 1/2, f(3) = 1/3, f(4) = 1/4, \dots$ So the sequence is 1, 1/2, 1/3, 1/4, ... with $a_n = 1/n$.

Exam tip: For "define a sequence", the key phrase is "a function whose domain is the set of natural numbers". Mention that it is a real sequence when the range is a subset of $\mathbb{R}$.

Ways of representing a sequence

Definition: A recursive relation gives the first few terms of a sequence and a rule expressing each later term through earlier ones, e.g. $a_{n+1} = a_n + a_{n-1}$.

ഒരു sequence-ന്റെ ആദ്യ ചില terms നൽകി, തുടർന്നുള്ള ഓരോ term-ഉം മുമ്പുള്ള terms ഉപയോഗിച്ച് എഴുതുന്ന rule ആണ് recursive relation; ഉദാ. $a_{n+1} = a_n + a_{n-1}$.

A sequence can be given in four ways: (a) by listing the first few terms until the rule is clear; (b) by a formula for the n th term a_n ; (c) by a **recursive relation**, giving the first few terms and a rule for getting later terms from earlier ones; (d) by more than one formula, for example one rule for odd n and another for even n .

മലയാളത്തിൽ

ഒരു sequence പറഞ്ഞുതരാൻ നാല് വഴികളുണ്ട്. ആദ്യത്തെ കുറച്ച് terms എഴുതി pattern കാണിക്കാം; a_n -ന് ഒരു formula തരാം; മുൻപത്തെ terms ഉപയോഗിച്ച് അടുത്തത് കണ്ടെത്തുന്ന recursive relation തരാം (Fibonacci sequence പോലെ); അല്ലെങ്കിൽ odd n -ന് ഒരു formula, even n -ന് വേറൊന്ന് എന്ന രീതിയിൽ തരാം. ചോദ്യത്തിൽ ഏത് രീതിയിൽ തന്നാലും, n -ന്റെ value substitute ചെയ്ത് terms കണ്ടുപിടിക്കുക എന്നതാണ് നമ്മുടെ ജോലി.

- (a) Listing: 1, 1/4, 1/9, 1/16, ... ($a_n = 1/n^2$); 5, 10, 15, 20, ... ($a_n = 5n$)
- (b) nth-term formula: $a_n = n^2$ gives 1, 4, 9, 16, ...; $a_n = n/(n+1)$ gives 1/2, 2/3, 3/4, 4/5, ...
- (c) Recursive (Fibonacci): $a_1 = a_2 = 1$, $a_{n+1} = a_n + a_{n-1}$ ($n \geq 2$) gives 1, 1, 2, 3, 5, 8, ...
- (d) Piecewise: $a_n = (n-1)/2$ if n is odd, $n/2$ if n is even, gives 0, 1, 1, 2, 2, 3, 3, ...

EXAMPLE 1 (I), (II)

- (i) $a_n = n^2 - 2n + 2$. Find a_1, a_2, a_r . (ii) $a_n = (1 + (-1)^n)/2$. Find a_1 to a_4 .
- (i) $a_1 = 1 - 2 + 2 = 1$; $a_2 = 4 - 4 + 2 = 2$; $a_r = r^2 - 2r + 2$ (just replace n by r).
- (ii) For odd n , $(-1)^n = -1$, so $a_n = 0/2 = 0$. For even n , $(-1)^n = 1$, so $a_n = 2/2 = 1$. Hence $a_1 = 0, a_2 = 1, a_3 = 0, a_4 = 1$.

EXAMPLE 1 (III) - RECURSIVE

- $a_1 = 2, a_2 = 3, a_n = 2a_{n-1} + 4a_{n-2}$ for $n \geq 3$. Find a_3, a_4 .
- $a_3 = 2a_2 + 4a_1 = 2 \times 3 + 4 \times 2 = 6 + 8 = 14$.
- $a_4 = 2a_3 + 4a_2 = 2 \times 14 + 4 \times 3 = 28 + 12 = 40$.

EXAMPLE 1 (IV), (V)

- (iv) $a_n = n^2$ for $n = 1, 3, 5, \dots$ and $a_n = 1/(n+1)$ for $n = 2, 4, 6, \dots$. Find a_1 to a_6 . (v) $a_n = (n-1)(n-2)/(2(n+1))$. Find a_1 to a_4 .
- (iv) Odd places use n^2 : $a_1 = 1, a_3 = 9, a_5 = 25$. Even places use $1/(n+1)$: $a_2 = 1/3, a_4 = 1/5, a_6 = 1/7$.
- (v) $a_1 = (0)(-1)/4 = 0$; $a_2 = (1)(0)/6 = 0$; $a_3 = (2)(1)/8 = 1/4$; $a_4 = (3)(2)/10 = 3/5$.

Exam tip: In a recursive sequence you cannot jump to a_4 directly; compute a_3 first. With piecewise rules, check whether n is odd or even before substituting.

Arithmetic Progression (A.P.)

Definition: An arithmetic progression is a sequence in which each term minus the preceding term is a constant d , the common difference: $a_{n+1} - a_n = d$ for all n .

ഓരോ term-ൽ നിന്നും തൊട്ടുമുമ്പുള്ള term കുറച്ചാൽ എപ്പോഴും ഒരേ constant d കിട്ടുന്ന sequence ആണ് arithmetic progression; d ആണ് common difference: എല്ലാ n -നും $a_{n+1} - a_n = d$.

A sequence $\langle a_n \rangle$ is an **arithmetic progression** if the difference between any term and the term before it is always the same constant d , called the **common difference**. For example, 7, 11, 15, 19, ... is an A.P. with first term 7 and $d = 4$. To test a list of terms, check $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots$. To test a formula a_n , compute $a_{n+1} - a_n$: if it does not depend on n , the sequence is an A.P.

മലയാളത്തിൽ

ഓരോ term-ഉം തൊട്ടുമുമ്പത്തെ term-ൽ നിന്ന് ഒരേ അളവിൽ കൂട്ടുകയോ കുറയുകയോ ചെയ്യുന്നുണ്ടെങ്കിൽ അത് A.P. ആണ്. എല്ലാ വർഷവും ₹1,500 fixed increment കിട്ടുന്ന ഒരു ശമ്പളം ആലോചിക്കുക: 25,000, 26,500, 28,000, ... ഇവിടെ common difference $d = 1500$. Formula തന്നാൽ $a_{n+1} - a_n$ കണക്കാക്കുക; ഉത്തരത്തിൽ n വരുന്നില്ലെങ്കിൽ A.P., n വന്നാൽ A.P. അല്ല.

A.P.: $a_{n+1} - a_n = d$ for all $n \in \mathbb{N}$ (d constant)

Test by terms: $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \dots = d$

Test by formula: $a_{n+1} - a_n$ must be independent of n

EXAMPLE 2 - TESTING LISTS

Which are A.P.s? (i) 1, $1/2$, $1/3$, $1/4$, ... (ii) 1, 4, 9, 16, ... (iii) 1, 4, 7, 10, ... (iv) 8, $7\frac{1}{3}$, $6\frac{2}{3}$, 6, ...

(i) $a_2 - a_1 = 1/2 - 1 = -1/2$; $a_3 - a_2 = 1/3 - 1/2 = -1/6$. Not equal, so not an A.P.

(ii) $4 - 1 = 3$; $9 - 4 = 5$. Not equal, so not an A.P.

(iii) $4 - 1 = 3$, $7 - 4 = 3$, $10 - 7 = 3$. A.P. with first term 1, $d = 3$.

(iv) $7\frac{1}{3} - 8 = 22/3 - 24/3 = -2/3$; $6\frac{2}{3} - 7\frac{1}{3} = 20/3 - 22/3 = -2/3$; $6 - 6\frac{2}{3} = 18/3 - 20/3 = -2/3$. A.P. with first term 8, $d = -2/3$.

EXAMPLE 2 - TESTING FORMULAS

(v) $a_n = 4n + 5$ (vi) $a_n = n^2 + n$

(v) $a_{n+1} = 4(n+1) + 5 = 4n + 9$. So $a_{n+1} - a_n = (4n + 9) - (4n + 5) = 4$, independent of n . A.P. with first term $a_1 = 4 + 5 = 9$ and $d = 4$.

(vi) $a_{n+1} = (n+1)^2 + (n+1) = n^2 + 3n + 2$. So $a_{n+1} - a_n = 2n + 2$, which depends on n . Not an A.P.

Exam tip: Checking just two differences is enough to prove a list is not an A.P., but to prove it is one from a formula, show $a_{n+1} - a_n$ is a constant for general n . Any a_n of the form $pn + q$ is an A.P. with $d = p$.

General term of an A.P.

Definition: The general (n th) term of an A.P. with first term a and common difference d is $T_n = a + (n - 1)d$.

First term a -യും common difference d -യും ഉള്ള A.P.-യുടെ general (n th) term $T_n = a + (n - 1)d$ ആണ്.

The **standard A.P.** is $a, a + d, a + 2d, a + 3d, \dots$ with first term a and common difference d . The 1st term has 0 d 's added, the 2nd has 1, the 3rd has 2, so the n th term has $(n - 1)$ d 's added. This gives the general term $T_n = a + (n - 1)d$. Use it to find any term, to find which term equals a given value, or to find the A.P. from two known terms.

മലയാളത്തിൽ

ശമ്പള ഉദാഹരണം തന്നെ എടുക്കാം: ആദ്യ വർഷം ₹25,000, ഓരോ വർഷവും ₹1,500 increment. 10-ാം വർഷം എത്ര? ആദ്യ വർഷത്തിന് ശേഷം 9 increment മാത്രമേ കിട്ടിയിട്ടുള്ളൂ, അതുകൊണ്ട് $25000 + 9 \times 1500 = ₹38,500$. ഇതാണ് $T_n = a + (n - 1)d$ -ന്റെ അർത്ഥം: n -ാം term-ൽ എത്രയോൾ $(n - 1)$ തവണ d കൂട്ടിയിട്ടുണ്ടാകും. ഒരു value ഏത് term ആണെന്ന് ചോദിച്ചാൽ $T_n = \text{value}$ എന്ന് വെച്ച് n കണ്ടുപിടിക്കുക; n ഒരു natural number അല്ലെങ്കിൽ ആ value sequence-ൽ ഇല്ല.

Standard A.P.: $a, a + d, a + 2d, a + 3d, \dots$

n th term: $T_n = a_n = a + (n - 1)d$

$$d = T_2 - T_1$$

EXAMPLE 3 (I)-(III)

(i) $a = 4, d = 3$. Find T_n and T_{17} . (ii) Find T_n of $5, 5 + \sqrt{3}, 5 + 2\sqrt{3}, 5 + 3\sqrt{3}, \dots$ (iii) Find T_{20} of $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$

(i) $T_n = 4 + (n - 1)3 = 3n + 1$. $T_{17} = 3 \times 17 + 1 = 52$.

(ii) $a = 5, d = (5 + \sqrt{3}) - 5 = \sqrt{3}$. $T_n = 5 + (n - 1)\sqrt{3}$.

(iii) $a = 20, d = 77/4 - 20 = -3/4$. $T_{20} = a + 19d = 20 - 57/4 = (80 - 57)/4 = 23/4 = 5\frac{3}{4}$.

EXAMPLE 3 (IV), (V) - WHICH TERM?

(iv) Which term of $43, 38, 33, 28, \dots$ is -457 ? (v) Which term of $17, 16\frac{1}{2}, 16, 15\frac{1}{2}, \dots$ is the first negative term?

(iv) $a = 43, d = -5$. Set $T_n = -457$: $43 + (n - 1)(-5) = -457 \Rightarrow 48 - 5n = -457 \Rightarrow 5n = 505 \Rightarrow n = 101$. So -457 is the 101st term.

(v) $a = 17, d = -1/2$. We need $T_n < 0$: $17 - (n - 1)/2 < 0 \Rightarrow (n - 1)/2 > 17 \Rightarrow n - 1 > 34 \Rightarrow n > 35$. The smallest natural number greater than 35 is 36. Check: $T_{35} = 17 - 17 = 0$ and $T_{36} = -1/2$. So the 36th term is the first negative term.

EXAMPLE 3 (VI) - A.P. FROM TWO TERMS

The 7th and 31st terms of an A.P. are 29 and 125. Find the A.P.

$T_7 = a + 6d = 29 \dots (1); T_{31} = a + 30d = 125 \dots (2)$.

$(2) - (1): 24d = 96 \Rightarrow d = 4$. From (1): $a + 24 = 29 \Rightarrow a = 5$.

The A.P. is $5, 9, 13, 17, \dots$

Exam tip: The most common slip is writing $a + nd$ instead of $a + (n - 1)d$. For "first negative term" questions, $T_n = 0$ is not negative, so check the boundary term.

Sum of n terms of an A.P.

Definition: The sum of the first n terms of an A.P. is $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$, where l is the last term.

ഒരു A.P.-യുടെ ആദ്യ n terms-ന്റെ sum $S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$ ആണ്; ഇവിടെ l ആണ് last term.

Let $S_n = a + (a + d) + \dots + [a + (n - 1)d]$. Write the same sum in reverse order and add the two lines term by term: each pair adds to $2a + (n - 1)d$, and there are n pairs. So $2S_n = n[2a + (n - 1)d]$. Since $a + (n - 1)d$ is the last term l , the formula can also be written as $S_n = (n/2)(a + l)$: number of terms times the average of the first and last terms.

മലയാളത്തിൽ

ഒരു RD അല്ലെങ്കിൽ chitty ആലോചിക്കുക: ആദ്യ മാസം ₹500 അടയ്ക്കുന്നു, ഓരോ മാസവും ₹100 കൂട്ടി അടയ്ക്കുന്നു. 12 മാസം കൊണ്ട് ആകെ എത്ര അടച്ചു? ഓരോ മാസവും കൂട്ടി നോക്കേണ്ട കാര്യമില്ല. ആദ്യത്തെയും അവസാനത്തെയും തുക കൂട്ടി 2 കൊണ്ട് ഹരിച്ചാൽ ശരാശരി മാസത്തുക കിട്ടും, അതിനെ മാസങ്ങളുടെ എണ്ണം കൊണ്ട് ഗുണിച്ചാൽ മതി. അതാണ് $S_n = (n/2)(a + l)$. Last term അറിയില്ലെങ്കിൽ $S_n = (n/2)[2a + (n - 1)d]$ ഉപയോഗിക്കുക.

$$S_n = (n/2)[2a + (n - 1)d]$$

$$S_n = (n/2)(a + l), \text{ where } l = a + (n - 1)d \text{ is the last term}$$

EXAMPLE 4

Find: (i) $1 + 4 + 7 + 10 + \dots$ to 40 terms (ii) $0.8 + 0.81 + 0.82 + \dots$ to 101 terms (iii) $3 + 7 + 11 + \dots + 79$

(i) $a = 1, d = 3, n = 40$. $S_{40} = 20[2 + 39 \times 3] = 20(2 + 117) = 20 \times 119 = 2380$.

(ii) $a = 0.8, d = 0.01, n = 101$. $S_{101} = (101/2)[1.6 + 100 \times 0.01] = (101/2)(2.6) = 101 \times 1.3 = 131.3$.

(iii) $a = 3, d = 4$, last term 79. First find n : $3 + (n - 1)4 = 79 \Rightarrow 4n - 1 = 79 \Rightarrow n = 20$. Then $S_{20} = (20/2)(3 + 79) = 10 \times 82 = 820$. (Check with the other form: $10[6 + 19 \times 4] = 10 \times 82 = 820$.)

EXAMPLE - CHITTY/RD

A person pays ₹500 in the first month and increases the payment by ₹100 every month. Find the payment in the 12th month and the total paid in 12 months.

$$a = 500, d = 100, n = 12. T_{12} = 500 + 11 \times 100 = ₹1,600.$$

$$S_{12} = (12/2)(500 + 1600) = 6 \times 2100 = ₹12,600.$$

Exam tip: When the question gives the last term (like " $+ \dots + 79$ ") but not n , find n first from $T_n = l$, then use $S_n = (n/2)(a + l)$.

Geometric Progression (G.P.) and its general term

Geometric progression: A sequence of non-zero terms in which each term divided by the preceding term is a constant $r \neq 0$, the common ratio.

ഓരോ term-നെയും തൊട്ടുമുമ്പുള്ള term കൊണ്ട് ഹരിച്ചാൽ എപ്പോഴും ഒരേ constant $r (\neq 0)$ കിട്ടുന്ന, non-zero terms ഉള്ള sequence ആണ് geometric progression; r ആണ് common ratio.

General term: $T_n = ar^{n-1}$, where a is the first term and r the common ratio.

$T_n = ar^{n-1}$; ഇവിടെ a first term-ഉം r common ratio-യും ആണ്.

A sequence $\langle a_n \rangle$ is a **geometric progression** if the ratio a_{n+1}/a_n of any term to the term before it is the same non-zero constant r , called the **common ratio**. In a G.P. no term can be zero and $r \neq 0$. The standard G.P. is $a, ar, ar^2, ar^3, \dots$; the n th term carries r to the power $(n - 1)$, so $T_n = ar^{n-1}$.

മലയാളത്തിൽ

A.P.-ൽ ഓരോ തവണയും ഒരു സംഖ്യ കൂട്ടുന്നു; G.P.-ൽ ഓരോ തവണയും ഒരു സംഖ്യ കൊണ്ട് ഗുണിക്കുന്നു. ബാങ്കിൽ ₹10,000 10% compound interest-ൽ ഇട്ടാൽ ഓരോ വർഷവും തുക 1.1 കൊണ്ട് ഗുണിക്കപ്പെടുന്നു: 11,000, 12,100, 13,310, ... ഇതൊരു G.P. ആണ്, $r = 1.1$. G.P. ആണോ എന്നറിയാൻ $T_2/T_1, T_3/T_2, \dots$ കണക്കാക്കി എല്ലാം തുല്യമാണോ എന്ന് നോക്കുക.

G.P.: $a_{n+1}/a_n = r$ for all $n \in \mathbb{N}$ ($r \neq 0, a_n \neq 0$)

Standard G.P.: $a, ar, ar^2, ar^3, \dots$

n th term: $T_n = ar^{n-1}$

$r = T_2/T_1$

EXAMPLE 5

Which are G.P.s? Give a, r and T_n where possible. (i) 2, 6, 18, 54, ... (ii) $1/8, -1/4, 1/2, -1, \dots$ (iii) $\sqrt{2}, 3\sqrt{2}, 6\sqrt{2}, 12\sqrt{2}, \dots$

(i) $6/2 = 18/6 = 54/18 = 3$. G.P. with $a = 2, r = 3, T_n = 2(3)^{n-1}$.

(ii) $(-1/4)/(1/8) = -2, (1/2)/(-1/4) = -2, (-1)/(1/2) = -2$. G.P. with $a = 1/8, r = -2, T_n = (1/8)(-2)^{n-1}$.

(iii) $3\sqrt{2}/\sqrt{2} = 3$ but $6\sqrt{2}/3\sqrt{2} = 2$. Ratios differ, so not a G.P.

EXAMPLE - COMPOUND INTEREST

₹10,000 is deposited at 10% per annum compounded yearly. Find the amount at the end of 3 years.

Amounts at the end of years 1, 2, 3, ... form a G.P. with first term $10000 \times 1.1 = 11000$ and $r = 1.1$.

End of year 3 = $T_3 = 11000 \times (1.1)^2 = 11000 \times 1.21 = ₹13,310$.

Exam tip: A negative r is allowed; the signs then alternate, as in (ii). When two terms of a G.P. are given (say T_4 and T_7), divide the equations to eliminate a .

Sum of n terms of a G.P.

Definition: The sum of the first n terms of a G.P. is $S_n = a(1 - r^n)/(1 - r) = a(r^n - 1)/(r - 1)$ for $r \neq 1$, and $S_n = na$ for $r = 1$.

ഒരു G.P.-യുടെ ആദ്യ n terms-ന്റെ sum, $r \neq 1$ ആണെങ്കിൽ $S_n = a(1 - r^n)/(1 - r) = a(r^n - 1)/(r - 1)$; $r = 1$ ആണെങ്കിൽ $S_n = na$.

Let $S_n = a + ar + \dots + ar^{n-1}$. Multiply by r to get $rS_n = ar + ar^2 + \dots + ar^n$. Subtracting, all middle terms cancel in pairs, leaving $(1 - r)S_n = a - ar^n$. So $S_n = a(1 - r^n)/(1 - r)$. Taking -1 common from top and bottom gives the equivalent form $a(r^n - 1)/(r - 1)$. Both give the same answer; use the first when $r < 1$ and the second when $r > 1$ so the numbers stay positive.

മലയാളത്തിൽ

G.P. sum-ന്റെ trick ഇതാണ്: S_n -നെ r കൊണ്ട് ഗുണിച്ചാൽ എല്ലാ terms-ഉം ഒരു സ്ഥാനം നീങ്ങുന്നു. രണ്ടും കുറച്ചാൽ ആദ്യത്തെ term a -യും അവസാനത്തെ ar^n -ഉം മാത്രം ബാക്കിയാകും. $r > 1$ ആണെങ്കിൽ $a(r^n - 1)/(r - 1)$, $r < 1$ ആണെങ്കിൽ $a(1 - r^n)/(1 - r)$ ഉപയോഗിക്കുക. $r = 1$ ആണെങ്കിൽ ഈ formula ഉപയോഗിക്കാൻ പറ്റില്ല: അപ്പോൾ എല്ലാ terms-ഉം a ആണ്, sum = na .

$$S_n = a(1 - r^n)/(1 - r), \text{ when } r < 1$$

$$S_n = a(r^n - 1)/(r - 1), \text{ when } r > 1$$

$$(\text{If } r = 1, S_n = na)$$

EXAMPLE 6

Find: (i) $2 + 4 + 8 + 16 + \dots$ to 10 terms (ii) $-1 + 2/3 - 4/9 + \dots$ to 8 terms

(i) $a = 2, r = 4/2 = 2 > 1, n = 10. S_{10} = 2(2^{10} - 1)/(2 - 1) = 2(1024 - 1) = 2046.$

(ii) $a = -1, r = (2/3)/(-1) = -2/3 < 1, n = 8. (-2/3)^8 = 256/6561.$

$S_8 = (-1)(1 - 256/6561)/(1 + 2/3) = -(6305/6561) \times (3/5) = -1261 \times 3/6561 = -3783/6561 = -1261/2187.$

EXAMPLE

Find $3 + 6 + 12 + \dots$ to 6 terms.

$a = 3, r = 2, n = 6. S_6 = 3(2^6 - 1)/(2 - 1) = 3 \times 63 = 189.$ (Check: $3 + 6 + 12 + 24 + 48 + 96 = 189.$)

Exam tip: In Example 6(i) the unit writes " $d = 4/2$ "; it should be $r = 4/2 = 2$ (a G.P. has a common ratio, not a common difference). With a negative r , put brackets: $(-2/3)^8$ is positive because the power is even.

Sum of an infinite G.P.

Definition: An infinite G.P. $a + ar + ar^2 + \dots$ has a finite sum only when $|r| < 1$, and then $S_\infty = a/(1 - r).$

Infinite G.P. $a + ar + ar^2 + \dots$ —finite sum ഉണ്ടാകുന്നത് $|r| < 1$ ആകുമ്പോൾ മാത്രമാണ്; അപ്പോൾ $S_\infty = a/(1 - r).$

A G.P. $a, ar, ar^2, \dots$ going on forever is an **infinite G.P.** Its sum is finite only when $|r| < 1$, i.e. $-1 < r < 1$. The reason: when $|r| < 1$, r^n gets closer and closer to 0 as $n \rightarrow \infty$ (for example 0.2, 0.04, 0.008, 0.0016, ...), so $S_n = a(1 - r^n)/(1 - r)$ approaches $a/(1 - r)$. When $r > 1$, r^n grows without bound (4, 16, 64, 256, ...) and there is no finite sum.

മലയാളത്തിൽ

ഒരു ഗ്യാസ് പായസത്തിന്റെ പകുതി കുടിക്കുന്നു, പിന്നെ ബാക്കിയുടെ പകുതി, പിന്നെ അതിന്റെ പകുതി... $1/2 + 1/4 + 1/8 + \dots$ എത്ര terms കൂട്ടിയാലും ആകെ ഒരു ഗ്യാസ് കവിയില്ല. $|r| < 1$ ആകുമ്പോൾ terms വളരെ വേഗം ചെറുതാകുന്നതുകൊണ്ട് infinite sum—ഉം finite ആയിരിക്കും, അത് $S = a/(1 - r)$. $|r| \geq 1$ ആണെങ്കിൽ ഈ formula ഉപയോഗിക്കരുത്.

$$S_{\infty} = a + ar + ar^2 + \dots = a/(1 - r), \text{ valid only for } |r| < 1$$

Reason: $\lim_{n \rightarrow \infty} r^n = 0$ when $|r| < 1$, so $\lim_{n \rightarrow \infty} S_n = a/(1 - r)$

Value of x	Behaviour of x^n as $n \rightarrow \infty$
$ x < 1$	$x^n \rightarrow 0$ (e.g. 0.2, 0.04, 0.008, ...)
$x = 1$	$x^n = 1$ for every n
$x > 1$	$x^n \rightarrow \infty$ (e.g. 4, 16, 64, 256, ...)
$x = -1$	oscillates: -1, 1, -1, 1, ...
$x < -1$	oscillates with growing size (e.g. -2, 4, -8, 16, ...)

EXAMPLE 7

Find: (i) $1 + 1/2 + 1/4 + 1/8 + \dots$ to ∞ (ii) $1 - 1/5 + 1/25 - 1/125 + \dots$ to ∞

(i) $a = 1, r = 1/2, |r| < 1. S = 1/(1 - 1/2) = 1/(1/2) = 2.$

(ii) $a = 1, r = -1/5, |r| < 1. S = 1/(1 + 1/5) = 1/(6/5) = 5/6.$

EXAMPLE - RECURRING DECIMAL

Write 0.454545... as a fraction.

$0.4545\dots = 45/100 + 45/10000 + \dots$, an infinite G.P. with $a = 45/100, r = 1/100.$

$S = (45/100)/(1 - 1/100) = (45/100)/(99/100) = 45/99 = 5/11.$

Exam tip: Always state " $|r| < 1$, so the sum exists" before using $a/(1 - r)$. The unit's Remark 8 refers to "Unit 5 of MST-001" for limits; that reference belongs to another course and can be ignored.

Concept of summation (Σ) and change of origin

Series: The expression $a_1 + a_2 + a_3 + \dots$ formed by adding the terms of a sequence, written compactly as Σa_n .

ഒരു sequence-ലെ terms കൂട്ടിയെഴുതുന്ന $a_1 + a_2 + a_3 + \dots$ എന്ന expression ആണ് series; ഇത് ചുരുക്കി Σa_n എന്ന് എഴുതുന്നു.

Dummy variable: The index letter under Σ ; replacing it by any other letter does not change the sum.

Σ -ന്റെ താഴെയുള്ള index letter ആണ് dummy variable; അതിന് പകരം മറ്റേത് letter ഉപയോഗിച്ചാലും sum മാറില്ല.

Change of origin: Rewriting a sum to start from a different index value: $\Sigma_{i=0}^n x_i = \Sigma_{i=k}^{n+k} x_{i-k}$.

ഒരു sum-നെ മറ്റൊരു index value-ൽ തുടങ്ങുന്ന രീതിയിൽ മാറ്റിയെഴുതുന്നതാണ് change of origin: $\Sigma_{i=0}^n x_i = \Sigma_{i=k}^{n+k} x_{i-k}$.

If $a_1, a_2, a_3, \dots$ is a sequence, the expression $a_1 + a_2 + a_3 + \dots$ is a **series**, written compactly with the Greek letter Σ (sigma). The letter below Σ ($n, i, k, \dots$) is a **dummy variable**: any letter can be used without changing the sum. The same series can be written with a different starting value (origin): to shift the origin from 0 to k , subtract k from the suffix inside, and for a finite sum add k to the upper limit.

മലയാളത്തിൽ

Σ ഒരു ചുരുക്കെഴുത്ത് മാത്രമാണ്: "താഴെ പറഞ്ഞ value മുതൽ മുകളിൽ പറഞ്ഞ value വരെ എല്ലാം കൂട്ടുക". i എന്നോ k എന്നോ എഴുതിയാലും sum മാറില്ല, അതുകൊണ്ടാണ് dummy variable എന്ന് പറയുന്നത്. Origin മാറ്റുന്നത് ഒരു ക്യൂവിൽ ടോക്കൺ നമ്പർ 0-ന് പകരം 2 മുതൽ തുടങ്ങുന്നത് പോലെയാണ്: ആളുകൾ മാറുന്നില്ല, നമ്പർ മാത്രം 2 കൂടുന്നു. അതുകൊണ്ട് അകത്ത് suffix-ൽ നിന്ന് 2 കുറയ്ക്കണം, അവസാന limit-ൽ 2 കൂട്ടണം.

$$\Sigma_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots \text{ to } \infty$$

$$\Sigma_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

$$\text{Infinite: } \Sigma_{n=0}^{\infty} x_n = \Sigma_{n=2}^{\infty} x_{n-2} = \Sigma_{n=10}^{\infty} x_{n-10} = \Sigma_{n=r}^{\infty} x_{n-r}$$

$$\text{Finite: } \Sigma_{i=0}^n x_i = \Sigma_{i=k}^{n+k} x_{i-k}$$

EXAMPLE

(a) Evaluate $\sum_{k=1}^4 (2k + 1)$. (b) Rewrite $\sum_{i=0}^3 x_i$ with origin at $i = 2$ and check it.

(a) $k = 1, 2, 3, 4$ give $3 + 5 + 7 + 9 = 24$.

(b) Subtract 2 from the suffix and add 2 to the upper limit: $\sum_{i=2}^5 x_{i-2}$. Check: $i = 2, 3, 4, 5$ give $x_0 + x_1 + x_2 + x_3$, the same four terms.

Exam tip: After shifting the origin, expand both forms for the first and last values of the index; they must give the same first and last terms. The unit prints the lower limit of the third form as "x = 10"; it should be $n = 10$.

Sums of some special sequences

Sum of natural numbers: $1 + 2 + \dots + n = n(n + 1)/2$.

ആദ്യ n natural numbers-ന്റെ sum: $1 + 2 + \dots + n = n(n + 1)/2$.

Sum of squares: $1^2 + 2^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$.

ആദ്യ n natural numbers-ന്റെ squares-ന്റെ sum: $1^2 + 2^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$.

Sum of cubes: $1^3 + 2^3 + \dots + n^3 = [n(n + 1)/2]^2$.

ആദ്യ n natural numbers-ന്റെ cubes-ന്റെ sum: $1^3 + 2^3 + \dots + n^3 = [n(n + 1)/2]^2$.

Three standard sums are used again and again: the sum of the first n natural numbers, of their squares, and of their cubes. The sum of cubes is exactly the square of the sum of the first n natural numbers. Sums like $\sum (k^2 + k)$ can be split into these standard pieces.

മലയാളത്തിൽ

ഈ മൂന്ന് formula-കളും മനഃപാഠമാക്കണം, പിന്നീടുള്ള units-ലും programme-ലുടനീളവും ഇവ വരും. ഓർക്കാൻ എളുപ്പമുള്ള ഒരു കാര്യം: cubes-ന്റെ sum = $(1 + 2 + \dots + n)^2$. $\sum (k^2 + k)$ പോലുള്ളവ കണ്ടാൽ $\sum k^2 + \sum k$ എന്ന് പിരിച്ച് ഓരോന്നിനും formula ഇടുക.

$$\sum_{k=1}^n k = 1 + 2 + \dots + n = n(n + 1)/2$$

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$$

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + \dots + n^3 = [n(n + 1)/2]^2 = (1 + 2 + \dots + n)^2$$

EXAMPLE

Find (i) $1 + 2 + \dots + 100$ (ii) $1^2 + 2^2 + \dots + 10^2$ (iii) $1^3 + 2^3 + \dots + 10^3$

(i) $100 \times 101/2 = 5050$.

(ii) $10 \times 11 \times 21/6 = 2310/6 = 385$.

(iii) $(10 \times 11/2)^2 = 55^2 = 3025$.

EXAMPLE

Find (i) $11^2 + 12^2 + \dots + 20^2$ (ii) $\sum_{k=1}^n k(k+1)$

(i) Sum up to 20 minus sum up to 10: $20 \times 21 \times 41/6 - 385 = 2870 - 385 = 2485$.

(ii) $\sum k(k+1) = \sum k^2 + \sum k = n(n+1)(2n+1)/6 + n(n+1)/2 = [n(n+1)/6][(2n+1) + 3] = n(n+1)(2n+4)/6 = n(n+1)(n+2)/3$. Check $n=2$: $1 \cdot 2 + 2 \cdot 3 = 8$ and $2 \cdot 3 \cdot 4/3 = 8$.

Exam tip: For a sum that starts in the middle (like 11 to 20), subtract the sum up to the term just before the start (up to 10, not 11).

Quick recap

- A sequence is a function with domain $\mathbb{N}$; a_n is its n th term. Real sequence: range $\subseteq \mathbb{R}$.
- A sequence can be given by listing terms, an n th-term formula, a recursive relation, or piecewise formulas.
- A.P.: $a_{n+1} - a_n = d$ (constant). From a formula, $a_{n+1} - a_n$ must be free of n .
- A.P. n th term: $T_n = a + (n-1)d$.
- If $T_n = \text{value}$ gives a non-natural n , the value is not a term of the sequence.
- A.P. sum: $S_n = (n/2)[2a + (n-1)d] = (n/2)(a+l)$.
- Three consecutive A.P. terms x, y, z satisfy $y - x = z - y$, i.e. $2y = x + z$.
- G.P.: $a_{n+1}/a_n = r$ (constant, non-zero); no term is zero.
- G.P. n th term: $T_n = ar^{n-1}$.
- G.P. sum: $a(1 - r^n)/(1 - r)$ for $r < 1$; $a(r^n - 1)/(r - 1)$ for $r > 1$.
- Infinite G.P. sum: $a/(1 - r)$, only when $|r| < 1$.
- $\sum$ uses a dummy variable; shifting the origin by k : subtract k in the suffix, add k to the upper limit.
- $\sum k = n(n+1)/2$, $\sum k^2 = n(n+1)(2n+1)/6$, $\sum k^3 = [n(n+1)/2]^2$.

Check yourself

Q. 1. If $a_n = \sqrt{2n}$, find a_1, a_2, a_3, a_4 . If $a_n = 2/\sqrt{n}$, find a_1, a_2, a_3, a_4, a_8 .

$$a_n = \sqrt{2n}: a_1 = \sqrt{2}, a_2 = \sqrt{4} = 2, a_3 = \sqrt{6}, a_4 = \sqrt{8} = 2\sqrt{2}.$$

$$a_n = 2/\sqrt{n}: a_1 = 2, a_2 = 2/\sqrt{2} = \sqrt{2}, a_3 = 2/\sqrt{3} = 2\sqrt{3}/3, a_4 = 2/2 = 1, a_8 = 2/(2\sqrt{2}) = 1/\sqrt{2} = \sqrt{2}/2.$$

Q. 2. If $a_n = 2^n \cdot 3^n / (2^n + 3^n)$, find a_1, a_2, a_3 .

$$a_1 = (2 \times 3)/(2 + 3) = 6/5.$$

$$a_2 = (4 \times 9)/(4 + 9) = 36/13.$$

$$a_3 = (8 \times 27)/(8 + 27) = 216/35.$$

Q. 3. Show that $\log a, \log(a^2/b), \log(a^3/b^2), \dots$ form an A.P.

$$\text{Use } \log m - \log n = \log(m/n).$$

$$a_2 - a_1 = \log(a^2/b) - \log a = \log(a^2/(ab)) = \log(a/b).$$

$$a_3 - a_2 = \log(a^3/b^2) - \log(a^2/b) = \log((a^3/b^2) \times (b/a^2)) = \log(a/b).$$

In general $a_n = \log(a^n/b^{n-1})$ and $a_{n+1} - a_n = \log(a/b)$, a constant. So it is an A.P. with $d = \log(a/b)$.

Q. 4. If $5k + 1, 6k + 5$ and $10k + 3$ are consecutive terms of an A.P., find k .

$$T_2 - T_1 = T_3 - T_2: (6k + 5) - (5k + 1) = (10k + 3) - (6k + 5).$$

$$k + 4 = 4k - 2 \Rightarrow 3k = 6 \Rightarrow k = 2.$$

Check: the terms are 11, 17, 23, with common difference 6.

Q. 5. Is 121 a term of the sequence 3, 9, 15, 21, ...?

$$a = 3, d = 6. \text{ Set } 3 + (n - 1)6 = 121 \Rightarrow 6n - 3 = 121 \Rightarrow 6n = 124 \Rightarrow n = 62/3 = 20\frac{2}{3}.$$

n must be a natural number, so 121 is not a term. ($T_{20} = 117$ and $T_{21} = 123$.)

Q. 6. How many terms are there in the A.P. $-1, -1/4, 1/2, 5/4, \dots, 14$?

$$a = -1, d = -1/4 - (-1) = 3/4.$$

$$-1 + (n - 1)(3/4) = 14 \Rightarrow (n - 1)(3/4) = 15 \Rightarrow n - 1 = 20 \Rightarrow n = 21.$$

There are 21 terms.

Q. 7. Find the 10th term of the G.P. 128, 32, 8, 2, ...

$$a = 128, r = 32/128 = 1/4.$$

$$T_{10} = 128 \times (1/4)^9 = 2^7/2^{18} = 1/2^{11} = 1/2048.$$

Q. 8. The 4th and 7th terms of a G.P. are 24 and 192. Find the G.P.

$$ar^3 = 24 \dots (1); ar^6 = 192 \dots (2).$$

$$(2) \div (1): r^3 = 8 \Rightarrow r = 2. \text{ From (1): } 8a = 24 \Rightarrow a = 3.$$

The G.P. is 3, 6, 12, 24, 48, ...

Q. 9. Find the sum $2/9 + 2/3 + 2 + 6 + \dots + 486$.

$$a = 2/9, r = (2/3)/(2/9) = 3.$$

$$\text{Find } n: (2/9) \cdot 3^{n-1} = 486 \Rightarrow 3^{n-1} = 486 \times 9/2 = 2187 = 3^7 \Rightarrow n = 8.$$

$$S_8 = (2/9)(3^8 - 1)/(3 - 1) = (2/9)(6560)/2 = 6560/9 = 728 \frac{8}{9}.$$

Q. 10. Prove that (i) $4^{1/4} \cdot 4^{1/8} \cdot 4^{1/16} \cdot 4^{1/32} \dots$ to $\infty = 2$ (ii) $5^{1/3} \cdot 5^{1/9} \cdot 5^{1/27} \dots$ to $\infty = \sqrt{5}$.

When powers of the same base are multiplied, the exponents add.

(i) Exponent = $1/4 + 1/8 + 1/16 + \dots$, an infinite G.P. with $a = 1/4, r = 1/2$ ($|r| < 1$). Sum = $(1/4)/(1 - 1/2) = 1/2$. So L.H.S. = $4^{1/2} = 2 = \text{R.H.S.}$

(ii) Exponent = $1/3 + 1/9 + 1/27 + \dots$ with $a = 1/3, r = 1/3$. Sum = $(1/3)/(2/3) = 1/2$. So L.H.S. = $5^{1/2} = \sqrt{5} = \text{R.H.S.}$

UNIT 6

Matrix Algebra

ഈ unit-ൽ matrix എന്താണ്, അതിന്റെ വിവിധ types, addition, multiplication, transpose, adjoint, inverse, rank എന്നിവ പഠിക്കാം. അവസാനം matrix method ഉപയോഗിച്ച് simultaneous equations എങ്ങനെ solve ചെയ്യാം എന്നും കാണാം.

Definition of a matrix

Row, column, rectangular, null and negative matrices

Diagonal, scalar, identity and triangular matrices

Comparable and equal matrices

Addition, subtraction and scalar multiplication

Matrix multiplication

Transpose of a matrix

Integral powers of a square matrix

Cofactors and the adjoint matrix

Trace of a matrix

Sub-matrices and minors

Inverse of a matrix

Orthogonal matrix

Symmetric and skew-symmetric matrices

Idempotent matrix

Partitioned matrices

Rank of a matrix

Application: solving linear equations by the matrix method

Definition of a matrix

Definition: A matrix is a rectangular arrangement of numbers (elements) in rows and columns; with m rows and n columns its order is $m \times n$, written $A = [a_{ij}]_{m \times n}$.

Rows-ലും columns-ലുമായി numbers (elements) ക്രമീകരിച്ച rectangular arrangement ആണ് matrix; m rows-ഉം n columns-ഉം ഉണ്ടെങ്കിൽ അതിന്റെ order $m \times n$ ആണ്, $A = [a_{ij}]_{m \times n}$ എന്ന് എഴുതുന്നു.

A matrix is a rectangular arrangement of numbers (elements) in rows and columns, enclosed in square brackets $[]$ or large parentheses $()$. A matrix with m rows and n columns is said to be of order (or dimension) $m \times n$, and it has $m \cdot n$ elements. Matrices are named with capital letters A, B, C , and the element in the i -th row and j -th column is written a_{ij} . So $A = [a_{ij}]_{m \times n}$.

മലയാളത്തിൽ

Matrix എന്നാൽ numbers-നെ rows-ഉം columns-ഉം ആയി ഒരു ചതുരത്തിൽ അടുക്കി വെച്ചതാണ്. സ്കൂളിലെ attendance register ഓർക്കുക: ഓരോ row-ഉം ഒരു കുട്ടി, ഓരോ column-ഉം ഒരു ദിവസം. Order എഴുതുമ്പോൾ എപ്പോഴും ആദ്യം rows, പിന്നെ columns ($m \times n$). a_{23} എന്നാൽ 2-ാം row-ഉം 3-ാം column-ഉം ചേരുന്ന സ്ഥലത്തെ element.

$$A = [a_{ij}]_{m \times n}, i = 1, 2, \dots, m \text{ (row number)}, j = 1, 2, \dots, n \text{ (column number)}$$

$$\text{Order} = (\text{number of rows}) \times (\text{number of columns})$$

$$\text{Number of elements} = m \times n$$

In AI/ML a dataset is stored exactly like this: each row is one sample and each column is one feature.

EXAMPLE

Kavita has 9 hundred-rupee notes, 4 fifty-rupee notes and 5 ten-rupee notes. Shrikant has 17, 6 and 1. Tanu has 8, 3 and 2. Write this as a matrix.

Put one person per row and one note type per column (Rs 100, Rs 50, Rs 10):

$$N = \begin{pmatrix} 9 & 4 & 5 \\ 17 & 6 & 1 \\ 8 & 3 & 2 \end{pmatrix}$$

This is a 3×3 matrix. Row 2 is Shrikant's money; column 1 is the number of Rs 100 notes each person has; $n_{21} = 17$.

EXAMPLE

$$\text{Write the order of } A = \begin{pmatrix} 9 & 7 & 8 & -3 & -8 \\ 4 & 3 & 6 & 1 & -10 \\ 10 & 12 & 15 & 2 & 5 \end{pmatrix} \text{ and the elements } a_{23}, a_{14}, a_{35}, a_{22}, a_{31}, a_{32}.$$

A has 3 rows and 5 columns, so its order is 3×5 .

$$a_{23} \text{ (row 2, column 3)} = 6; a_{14} = -3; a_{35} = 5; a_{22} = 3; a_{31} = 10; a_{32} = 12.$$

Exam tip: a_{ij} is always row i , column j . (One line of the unit says "j-th row and i-th column"; that is a slip. Remark 1 in the same unit gives the correct rule: the first suffix is the row.)

Row, column, rectangular, null and negative matrices

Row matrix: A matrix with only one row (order $1 \times n$).

ഒരു row മാത്രമുള്ള matrix (order $1 \times n$).

Column matrix: A matrix with only one column (order $m \times 1$).

ഒരു column മാത്രമുള്ള matrix (order $m \times 1$).

Rectangular matrix: An $m \times n$ matrix with $m \neq n$; if $m = n$ it is a square matrix.

$m \neq n$ ആയ $m \times n$ matrix; $m = n$ ആണെങ്കിൽ അത് square matrix ആണ്.

Null (zero) matrix: A matrix every element of which is 0, written O.

എല്ലാ elements-ഉം 0 ആയ matrix; O എന്ന് എഴുതുന്നു.

Negative matrix: $-A = [-a_{ij}]$, obtained by changing the sign of every element of A.

A-യിലെ ഓരോ element-ന്റെയും sign മാറ്റി കിട്ടുന്ന matrix $-A = [-a_{ij}]$.

Matrices are classified by their shape and by their elements. A **row matrix** has only one row (order $1 \times n$); a **column matrix** (column vector) has only one column (order $m \times 1$). A matrix with $m \neq n$ is a **rectangular matrix**; if $m = n$ it is a **square matrix** of order n. A **null (zero) matrix** $O_{m \times n}$ has every element 0. The **negative** of A, written $-A$, changes the sign of every element.

മലയാളത്തിൽ

ഒരു row മാത്രം ഉള്ളത് row matrix, ഒരു column മാത്രം ഉള്ളത് column matrix. Rows-ഉം columns-ഉം തുല്യമാണെങ്കിൽ square matrix, അല്ലെങ്കിൽ rectangular. എല്ലാ elements-ഉം പൂജ്യമായാൽ null matrix, അത് സാധാരണ algebra-യിലെ 0 പോലെ പെരുമാറും. $-A$ കിട്ടാൻ ഓരോ element-ന്റെയും sign മാറ്റിയാൽ മതി.

Row matrix: $1 \times n$, e.g. $\begin{pmatrix} 2 & 5 & 7 \end{pmatrix}$

Column matrix: $m \times 1$, e.g. $\begin{pmatrix} 9 \\ -3 \\ 2 \end{pmatrix}$

Square: $m = n$; Rectangular: $m \neq n$

Null matrix $O_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

If $A = [a_{ij}]$ then $-A = [-a_{ij}]$

EXAMPLE

Write all possible orders of a matrix having 8 elements.

The order $m \times n$ must satisfy $m \cdot n = 8$. The factor pairs of 8 are 1·8, 2·4, 4·2, 8·1.

So the possible orders are 1×8 , 2×4 , 4×2 and 8×1 .

EXAMPLE

Construct $A = [a_{ij}]_{3 \times 2}$ where $a_{ij} = |i - j|$.

$a_{11} = |1 - 1| = 0$, $a_{12} = |1 - 2| = 1$, $a_{21} = |2 - 1| = 1$, $a_{22} = |2 - 2| = 0$, $a_{31} = |3 - 1| = 2$, $a_{32} = |3 - 2| = 1$.

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 1 \end{pmatrix}$$

EXAMPLE

Find $-A$ if $A = \begin{pmatrix} 10 & 5 & 5 \\ -4 & -8 & 6 \end{pmatrix}$.

Change the sign of every element: $-A = \begin{pmatrix} -10 & -5 & -5 \\ 4 & 8 & -6 \end{pmatrix}$.

Exam tip: A 1×1 matrix such as $[6]$ is both a row matrix and a column matrix. The unit prints $O_{2 \times 3}$ with three rows; a 2×3 null matrix has 2 rows and 3 columns, as shown above.

Diagonal, scalar, identity and triangular matrices

Diagonal matrix: A square matrix in which every non-diagonal element is zero ($a_{ij} = 0$ for $i \neq j$).

Principal diagonal-ന് പുറത്തുള്ള എല്ലാ elements-ഉം zero ആയ square matrix ($i \neq j$ ആകുമ്പോൾ $a_{ij} = 0$).

Scalar matrix: A diagonal matrix whose diagonal elements are all equal.

Diagonal elements എല്ലാം തുല്യമായ diagonal matrix.

Identity matrix: A scalar matrix I_n whose diagonal elements are all 1.

Diagonal elements എല്ലാം 1 ആയ scalar matrix I_n .

Triangular matrix: A square matrix whose elements below the diagonal are all zero (upper triangular) or above the diagonal are all zero (lower triangular).

Diagonal-ന് താഴെയുള്ള elements എല്ലാം zero ആയ square matrix upper triangular; diagonal-ന് മുകളിലുള്ളവ എല്ലാം zero ആണെങ്കിൽ lower triangular.

In a square matrix, the elements $a_{11}, a_{22}, \dots, a_{nn}$ form the **principal diagonal**. A **diagonal matrix** has every non-diagonal element zero ($a_{ij} = 0$ for $i \neq j$); diagonal elements may be anything, even 0. A **scalar matrix** is a diagonal matrix whose diagonal elements are all equal. An **identity (unit) matrix** I_n is a scalar matrix with every diagonal element 1. An **upper triangular** matrix has all elements below the diagonal zero; a **lower triangular** matrix has all elements above the diagonal zero.

മലയാളത്തിൽ

Square matrix-ന്റെ മുകളിൽ ഇടത്തു നിന്ന് താഴെ വലത്തോട്ട് പോകുന്ന വരയാണ് principal diagonal. ആ വരയ്ക്ക് പുറത്ത് എല്ലാം പൂജ്യമാണെങ്കിൽ diagonal matrix.

Diagonal-ൽ എല്ലാം ഒരേ സംഖ്യയാണെങ്കിൽ scalar matrix, ആ സംഖ്യ 1 ആണെങ്കിൽ identity matrix I . Diagonal-ന് താഴെ എല്ലാം പൂജ്യമായാൽ upper triangular, മുകളിൽ എല്ലാം പൂജ്യമായാൽ lower triangular. ഒരു staircase പോലെ ചിന്തിച്ചാൽ എളുപ്പം ഓർക്കാം.

Diagonal: $a_{ij} = 0$ for all $i \neq j$, e.g. $\text{diag}[4, 3, 6] = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$

Scalar: diagonal with $a_{11} = a_{22} = \dots = a_{nn} = k$

Identity: $a_{ij} = 1$ if $i = j$, 0 if $i \neq j$; $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Upper triangular: $a_{ij} = 0$ for $i > j$; Lower triangular: $a_{ij} = 0$ for $i < j$

EXAMPLE

Write all diagonal matrices of order 2×2 whose elements are only 0 or 1.

Non-diagonal elements must be 0. The two diagonal places can be filled as (0, 0), (0, 1), (1, 0), (1, 1):

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

EXAMPLE

Write the order and type of: (i) $\begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$ (ii) $\begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$ (iii) $\begin{pmatrix} 2 & 5 & 7 \\ 0 & 8 & 0 \\ 0 & 0 & 9 \end{pmatrix}$ (iv) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 7 & 6 \end{pmatrix}$ (v) $\begin{pmatrix} 2 & 9 & 7 \\ 0 & 5 & 8 \\ 2 & 0 & 9 \end{pmatrix}$

(i) 2×2 diagonal (off-diagonal elements are 0).

(ii) 2×2 scalar (diagonal, and both diagonal elements equal 8).

(iii) 3×3 upper triangular (everything below the diagonal is 0).

(iv) 3×3 lower triangular (everything above the diagonal is 0).

(v) 3×3 square, but not upper triangular, because $a_{31} = 2 \neq 0$ lies below the diagonal.

Exam tip: Every identity matrix is scalar, and every scalar matrix is diagonal, but not the other way round. A diagonal matrix is both upper and lower triangular.

Comparable and equal matrices

Comparable matrices: Two matrices of the same order.

ഒരേ order ഉള്ള രണ്ട് matrices.

Equal matrices: Comparable matrices whose corresponding elements are all equal: $a_{ij} = b_{ij}$ for every i, j .

Corresponding elements എല്ലാം തുല്യമായ comparable matrices: എല്ലാ i, j -നും $a_{ij} = b_{ij}$.

Two matrices are **comparable** if they have the same order. They are **equal** if they are comparable and every pair of corresponding elements is equal. Equality of matrices therefore gives one ordinary equation for each position, which lets us find unknowns.

മലയാളത്തിൽ

ഒരേ order ഉള്ള matrices comparable ആണ്. ഒരേ order-ഉം, ഓരോ സ്ഥാനത്തെയും elements ഒന്നു തന്നെയും ആണെങ്കിൽ മാത്രമേ അവ equal ആകൂ. അതുകൊണ്ട് $A = B$ എന്ന് തന്നാൽ, ഓരോ position-ഉം compare ചെയ്ത് x, y, z പോലുള്ള unknowns കണ്ടുപിടിക്കാം.

$$A = [a_{ij}]_{m \times n}, B = [b_{ij}]_{m \times n}: A = B \Leftrightarrow a_{ij} = b_{ij} \text{ for every } i, j$$

EXAMPLE

$$\text{If } \begin{pmatrix} 3 & x+y \\ xy & 7+z \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 8 & 4 \end{pmatrix}, \text{ find } x, y, z.$$

Compare positions: $x + y = 6 \dots (1)$, $xy = 8 \dots (2)$, $7 + z = 4 \Rightarrow z = -3$.

From (1), $y = 6 - x$. Put in (2): $x(6 - x) = 8 \Rightarrow x^2 - 6x + 8 = 0 \Rightarrow (x - 4)(x - 2) = 0 \Rightarrow x = 4$ or 2 .

So $x = 4, y = 2, z = -3$, or $x = 2, y = 4, z = -3$.

EXAMPLE

$$\text{Find } x, y, z, w \text{ if } \begin{pmatrix} 3x - 2y & z + w \\ 3z - w & x + y \end{pmatrix} = \begin{pmatrix} -1 & 7 \\ 5 & 3 \end{pmatrix}.$$

$3x - 2y = -1 \dots (1)$, $z + w = 7 \dots (2)$, $3z - w = 5 \dots (3)$, $x + y = 3 \dots (4)$.

$(1) + 2 \times (4)$: $3x - 2y + 2x + 2y = -1 + 6 \Rightarrow 5x = 5 \Rightarrow x = 1$, and then $y = 3 - 1 = 2$.

$(2) + (3)$: $4z = 12 \Rightarrow z = 3$, and then $w = 7 - 3 = 4$.

$x = 1, y = 2, z = 3, w = 4$.

EXAMPLE

$$\text{If } \begin{pmatrix} a+5 & -2a & b+6 \\ 2c & 3b+2 & x \\ y+1 & z+3 & x+2 \end{pmatrix} = \begin{pmatrix} 2 & a+9 & 11 \\ c+4 & -b+22 & 3-x \\ 2 & 2z+3 & 5-x \end{pmatrix}, \text{ find } a, b, c, x, y, z.$$

$a + 5 = 2 \Rightarrow a = -3$ (check: $-2a = 6$ and $a + 9 = 6 \checkmark$). $b + 6 = 11 \Rightarrow b = 5$ (check: $3b + 2 = 17 = -b + 22 \checkmark$). $2c = c + 4 \Rightarrow c = 4$.

$x = 3 - x \Rightarrow x = 3/2$ (check: $x + 2 = 7/2 = 5 - x \checkmark$). $y + 1 = 2 \Rightarrow y = 1$. $z + 3 = 2z + 3 \Rightarrow z = 0$.

Addition, subtraction and scalar multiplication

Addition / subtraction: For matrices of the same order, $A \pm B = [a_{ij} \pm b_{ij}]$, formed by adding or subtracting corresponding elements.

ഒരേ order ഉള്ള matrices-ന്, corresponding elements കൂട്ടുകയോ കുറയ്ക്കുകയോ ചെയ്ത് കിട്ടുന്നതാണ് $A \pm B = [a_{ij} \pm b_{ij}]$.

Scalar multiplication: $kA = [k \cdot a_{ij}]$, formed by multiplying every element of A by the scalar k.

A-യിലെ ഓരോ element-നെയും scalar k കൊണ്ട് ഗുണിച്ച് കിട്ടുന്നതാണ് $kA = [k \cdot a_{ij}]$.

Two matrices can be added or subtracted only if they have the same order: add or subtract corresponding elements, and the result has the same order. To multiply a matrix by a scalar k, multiply every element by k. Addition is commutative and associative, O acts like zero, and $-A$ is the additive inverse of A. Division of one matrix by another is not defined.

മലയാളത്തിൽ

Addition-ഉം subtraction-ഉം ഒരേ order ഉള്ള matrices തമ്മിൽ മാത്രം. ഒരേ സ്ഥാനത്തുള്ള elements കൂട്ടുകയോ കുറയ്ക്കുകയോ ചെയ്താൽ മതി, രണ്ട് കടകളിലെ stock ഒരുമിച്ച് കൂട്ടുന്നതു പോലെ. kA എന്നാൽ ഓരോ element-നെയും k കൊണ്ട് ഗുണിക്കുക. Matrix-നെ matrix കൊണ്ട് ഹരിക്കുന്ന പരിപാടി ഇല്ല.

$$A + B = [a_{ij} + b_{ij}], A - B = [a_{ij} - b_{ij}] = A + (-1)B$$

$$kA = [k \cdot a_{ij}]$$

$$A + B = B + A; (A + B) + C = A + (B + C)$$

$$A + O = O + A = A; A + (-A) = O$$

EXAMPLE

If $A = \begin{pmatrix} 2 & 3 & 4 \\ 7 & 5 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 9 & 8 \end{pmatrix}$, find $A + B$ and $A - B$.

$$A + B = \begin{pmatrix} 2+1 & 3+5 & 4+6 \\ 7+2 & 5+9 & 1+8 \end{pmatrix} = \begin{pmatrix} 3 & 8 & 10 \\ 9 & 14 & 9 \end{pmatrix}$$

$$A - B = \begin{pmatrix} 2-1 & 3-5 & 4-6 \\ 7-2 & 5-9 & 1-8 \end{pmatrix} = \begin{pmatrix} 1 & -2 & -2 \\ 5 & -4 & -7 \end{pmatrix}$$

But if $A = \begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix}$ (2×2) and $B = \begin{pmatrix} 3 & 6 & 9 \\ 2 & 4 & 5 \end{pmatrix}$ (2×3), $A + B$ is not defined: the orders differ.

EXAMPLE

$A = \begin{pmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{pmatrix}$. Find $3A + 2B$ and $2A - 3B$.

$$3A = \begin{pmatrix} 6 & 12 & 15 \\ -9 & 18 & 21 \\ 3 & 24 & 27 \end{pmatrix}, 2B = \begin{pmatrix} 6 & 12 & 4 \\ 2 & 8 & 10 \\ 16 & 14 & -2 \end{pmatrix}$$

$$3A + 2B = \begin{pmatrix} 6+6 & 12+12 & 15+4 \\ -9+2 & 18+8 & 21+10 \\ 3+16 & 24+14 & 27-2 \end{pmatrix} = \begin{pmatrix} 12 & 24 & 19 \\ -7 & 26 & 31 \\ 19 & 38 & 25 \end{pmatrix}$$

$$2A = \begin{pmatrix} 4 & 8 & 10 \\ -6 & 12 & 14 \\ 2 & 16 & 18 \end{pmatrix}, 3B = \begin{pmatrix} 9 & 18 & 6 \\ 3 & 12 & 15 \\ 24 & 21 & -3 \end{pmatrix}$$

$$2A - 3B = \begin{pmatrix} 4-9 & 8-18 & 10-6 \\ -6-3 & 12-12 & 14-15 \\ 2-24 & 16-21 & 18+3 \end{pmatrix} = \begin{pmatrix} -5 & -10 & 4 \\ -9 & 0 & -1 \\ -22 & -5 & 21 \end{pmatrix}$$

EXAMPLE

Find X and Y if $3X + 2Y = \begin{pmatrix} 4 & 13 \\ 18 & 13 \end{pmatrix}$... (1) and $2X - 3Y = \begin{pmatrix} 7 & 0 \\ -1 & -13 \end{pmatrix}$... (2).

Eliminate Y like ordinary simultaneous equations: $3 \times (1) + 2 \times (2)$ gives $13X$.

$$13X = \begin{pmatrix} 12 & 39 \\ 54 & 39 \end{pmatrix} + \begin{pmatrix} 14 & 0 \\ -2 & -26 \end{pmatrix} = \begin{pmatrix} 26 & 39 \\ 52 & 13 \end{pmatrix} \Rightarrow X = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$$

$$\text{From (1): } 2Y = \begin{pmatrix} 4 & 13 \\ 18 & 13 \end{pmatrix} - 3X = \begin{pmatrix} 4 - 6 & 13 - 9 \\ 18 - 12 & 13 - 3 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 6 & 10 \end{pmatrix} \Rightarrow Y = \begin{pmatrix} -1 & 2 \\ 3 & 5 \end{pmatrix}$$

$$\text{Check (2): } 2X - 3Y = \begin{pmatrix} 4 + 3 & 6 - 6 \\ 8 - 9 & 2 - 15 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ -1 & -13 \end{pmatrix} \checkmark$$

Matrix multiplication

Definition: If A is $m \times n$ and B is $n \times p$, AB is the $m \times p$ matrix whose (i, k) element is $\sum_j a_{ij}b_{jk}$ (row i of A times column k of B).

A $m \times n$ -ഉം B $n \times p$ -ഉം ആണെങ്കിൽ AB ഒരു $m \times p$ matrix ആണ്; അതിന്റെ (i, k) element $\sum_j a_{ij}b_{jk}$ ആണ് (A-യുടെ row i-യും B-യുടെ column k-യും തമ്മിലുള്ള ഗുണനം).

The product AB is defined only when the number of columns of A equals the number of rows of B (A and B are "conformable"). If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$. The (i, j) element of AB is found by moving along row i of A and down column j of B, multiplying matching elements and adding. Matrix multiplication is associative and distributive, but in general **not commutative**: $AB \neq BA$.

മലയാളത്തിൽ

AB കിട്ടണമെങ്കിൽ A-യുടെ columns-ന്റെ എണ്ണം B-യുടെ rows-ന്റെ എണ്ണത്തിന് തുല്യമാകണം. $(m \times n)(n \times p)$ എന്നെഴുതിയാൽ നടുവിലെ രണ്ടും ഒന്നാകണം, ഉത്തരം പുറത്തെ രണ്ടിന്റെ order $m \times p$. A-യുടെ ഒരു row എടുത്ത് B-യുടെ ഒരു column-ഉമായി ഓരോ ജോടിയും ഗുണിച്ച് കൂട്ടുക; ഇതാണ് row-by-column rule. AB-യും BA-യും സാധാരണ ഒന്നല്ല, അതുകൊണ്ട് order മാറ്റി എഴുതരുത്.

$$A = [a_{ij}]_{m \times n}, B = [b_{jk}]_{n \times p} \Rightarrow AB = [c_{ik}]_{m \times p}, c_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$$

$$(AB)C = A(BC); A(B + C) = AB + AC; (A + B)C = AC + BC$$

$$I_n A = A I_n = A \text{ (A square of order } n)$$

$$\text{In general } AB \neq BA$$

In AI/ML, a neural-network layer computes $y = Wx$: the weight matrix W multiplied by the input vector x .

EXAMPLE

$$\text{Find } \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 5 & 2 \\ 9 & 7 \end{pmatrix}.$$

Orders: $(2 \times 3)(3 \times 2)$, inner numbers match, so the product exists and is 2×2 .

Row 1 $\times$ Column 1: $(1)(1) + (2)(5) + (3)(9) = 1 + 10 + 27 = 38$

Row 1 $\times$ Column 2: $(1)(3) + (2)(2) + (3)(7) = 3 + 4 + 21 = 28$

Row 2 $\times$ Column 1: $(0)(1) + (1)(5) + (2)(9) = 0 + 5 + 18 = 23$

Row 2 $\times$ Column 2: $(0)(3) + (1)(2) + (2)(7) = 0 + 2 + 14 = 16$

$$\text{Product} = \begin{pmatrix} 38 & 28 \\ 23 & 16 \end{pmatrix}$$

EXAMPLE

$$A = \begin{pmatrix} 5 & 4 \\ 2 & 7 \end{pmatrix}, B = \begin{pmatrix} 6 & 8 \\ 2 & 9 \end{pmatrix}. \text{ Show that } AB \neq BA.$$

AB: Row 1 $\times$ Col 1 = $5 \cdot 6 + 4 \cdot 2 = 38$; Row 1 $\times$ Col 2 = $5 \cdot 8 + 4 \cdot 9 = 76$; Row 2 $\times$ Col 1 = $2 \cdot 6 + 7 \cdot 2 = 26$; Row 2 $\times$ Col 2 = $2 \cdot 8 + 7 \cdot 9 = 79$.

$$AB = \begin{pmatrix} 38 & 76 \\ 26 & 79 \end{pmatrix}$$

BA: Row 1 $\times$ Col 1 = $6 \cdot 5 + 8 \cdot 2 = 46$; Row 1 $\times$ Col 2 = $6 \cdot 4 + 8 \cdot 7 = 80$; Row 2 $\times$ Col 1 = $2 \cdot 5 + 9 \cdot 2 = 28$; Row 2 $\times$ Col 2 = $2 \cdot 4 + 9 \cdot 7 = 71$.

$$BA = \begin{pmatrix} 46 & 80 \\ 28 & 71 \end{pmatrix} \neq AB.$$

Sometimes $AB = BA$ does happen: for $A = \begin{pmatrix} 2 & -3 \\ 4 & 5 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 3 \\ -4 & 2 \end{pmatrix}$, both products equal $\begin{pmatrix} 22 & 0 \\ 0 & 22 \end{pmatrix}$.

EXAMPLE

$$A = \begin{pmatrix} 2 & 4 & 5 \\ -3 & 6 & 7 \\ 1 & 8 & 9 \end{pmatrix}, B = \begin{pmatrix} 3 & 6 & 2 \\ 1 & 4 & 5 \\ 8 & 7 & -1 \end{pmatrix}. \text{ Find } AB.$$

Columns of B: $C1 = (3, 1, 8)$, $C2 = (6, 4, 7)$, $C3 = (2, 5, -1)$.

Row 1 $(2, 4, 5)$: $\cdot C1 = 6 + 4 + 40 = 50$; $\cdot C2 = 12 + 16 + 35 = 63$; $\cdot C3 = 4 + 20 - 5 = 19$

Row 2 $(-3, 6, 7)$: $\cdot C1 = -9 + 6 + 56 = 53$; $\cdot C2 = -18 + 24 + 49 = 55$; $\cdot C3 = -6 + 30 - 7 = 17$

Row 3 $(1, 8, 9)$: $\cdot C1 = 3 + 8 + 72 = 83$; $\cdot C2 = 6 + 32 + 63 = 101$; $\cdot C3 = 2 + 40 - 9 = 33$

$$AB = \begin{pmatrix} 50 & 63 & 19 \\ 53 & 55 & 17 \\ 83 & 101 & 33 \end{pmatrix}$$

EXAMPLE

Find AB, if defined: (i) $A = \begin{pmatrix} 5 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ (ii) $A = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \end{pmatrix}$ (iii) $A = \begin{pmatrix} 3 & 4 & 1 \\ 2 & 5 & 6 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$

(i) $(1 \times 2)(3 \times 1)$: $2 \neq 3$, so AB is not defined.

(ii) $(2 \times 1)(1 \times 2)$: defined, 2×2 . $AB = \begin{pmatrix} 3 \cdot 5 & 3 \cdot 6 \\ 4 \cdot 5 & 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 15 & 18 \\ 20 & 24 \end{pmatrix}$

(iii) $(2 \times 3)(3 \times 2)$: defined, 2×2 . Row 1 $\times$ Col 1 = $3 + 12 + 5 = 20$; Row 1 $\times$ Col 2 = $6 + 16 + 6 = 28$; Row 2 $\times$

Col 1 = $2 + 15 + 30 = 47$; Row 2 $\times$ Col 2 = $4 + 20 + 36 = 60$. $AB = \begin{pmatrix} 20 & 28 \\ 47 & 60 \end{pmatrix}$

EXAMPLE

Evaluate $\begin{pmatrix} 2 & 3 & 5 \end{pmatrix} \begin{pmatrix} 4 & 5 \\ 0 & 1 \\ 2 & 6 \end{pmatrix} \begin{pmatrix} 2 & 4 & 1 \\ 5 & 6 & 8 \end{pmatrix}$.

First product $(1 \times 3)(3 \times 2) = 1 \times 2$: $[2 \cdot 4 + 3 \cdot 0 + 5 \cdot 2, 2 \cdot 5 + 3 \cdot 1 + 5 \cdot 6] = \begin{pmatrix} 18 & 43 \end{pmatrix}$

Then $(1 \times 2)(2 \times 3) = 1 \times 3$: $[18 \cdot 2 + 43 \cdot 5, 18 \cdot 4 + 43 \cdot 6, 18 \cdot 1 + 43 \cdot 8] = [36 + 215, 72 + 258, 18 + 344] = \begin{pmatrix} 251 & 330 & 362 \end{pmatrix}$

Exam tip: Always write the orders first and check the inner numbers. If A is 3×2 and B is 2×4 , AB (3×4) exists but BA does not. If A is 3×2 and B is 2×3 , AB is 3×3 while BA is 2×2 , so they cannot be equal.

Transpose of a matrix

Definition: The transpose A^T (or A') of an $m \times n$ matrix A is the $n \times m$ matrix obtained by interchanging its rows and columns: $A^T = [a_{ji}]$.

$m \times n$ matrix A -യുടെ rows-ഉം columns-ഉം പരസ്പരം മാറ്റി കിട്ടുന്ന $n \times m$ matrix ആണ് transpose A^T (അഥവാ A'): $A^T = [a_{ji}]$.

The transpose of an $m \times n$ matrix A , written A' or A^T , is the $n \times m$ matrix obtained by interchanging rows and columns: row i of A becomes column i of A^T . Transposing twice gives back A . The transpose of a product reverses the order: $(AB)^T = B^T A^T$.

മലയാളത്തിൽ

Transpose എന്നാൽ rows-നെ columns ആക്കുക. ഒന്നാം row ഒന്നാം column ആകും, രണ്ടാം row രണ്ടാം column ആകും. 2×3 matrix-ന്റെ transpose 3×2 ആയിരിക്കും. Product-ന്റെ transpose എടുക്കുമ്പോൾ order തിരിയും: ആദ്യം socks, പിന്നെ shoes ഇട്ടാൽ, ഊരുമ്പോൾ ആദ്യം shoes, പിന്നെ socks.

$$\begin{aligned} A = [a_{ij}]_{m \times n} &\Rightarrow A^T = [a_{ji}]_{n \times m} \\ (A^T)^T &= A; (kA)^T = kA^T \\ (A \pm B)^T &= A^T \pm B^T \\ (AB)^T &= B^T A^T; (ABC)^T = C^T B^T A^T \end{aligned}$$

EXAMPLE

$$\text{If } A = \begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \end{pmatrix}, \text{ find } A^T.$$

Row 1 (2, 3, 4) becomes column 1; row 2 (5, 6, 8) becomes column 2: $A^T = \begin{pmatrix} 2 & 5 \\ 3 & 6 \\ 4 & 8 \end{pmatrix}$ (order 3×2).

EXAMPLE

$$A = \begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}, B = \begin{pmatrix} 6 & 8 \\ 10 & -12 \end{pmatrix}, C = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}. \text{ Verify } (A - B + C)^T = A^T - B^T + C^T.$$

$$A - B + C = \begin{pmatrix} 2 - 6 + 0 & -1 - 8 - 1 \\ 0 - 10 + 0 & -1 + 12 + 0 \end{pmatrix} = \begin{pmatrix} -4 & -10 \\ -10 & 11 \end{pmatrix}, \text{ so } (A - B + C)^T = \begin{pmatrix} -4 & -10 \\ -10 & 11 \end{pmatrix}.$$

$$A^T - B^T + C^T = \begin{pmatrix} 2 & 0 \\ -1 & -1 \end{pmatrix} - \begin{pmatrix} 6 & 10 \\ 8 & -12 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -4 & -10 \\ -10 & 11 \end{pmatrix} \checkmark$$

EXAMPLE

$$A = \begin{pmatrix} 3 & 0 \\ -4 & -1 \end{pmatrix}, B = \begin{pmatrix} 3 & 5 & -7 \\ 0 & -1 & 8 \end{pmatrix}, C = \begin{pmatrix} 6 \\ -1 \\ 0 \end{pmatrix}. \text{ Verify } (ABC)^T = C^T B^T A^T.$$

$$BC: \text{ Row 1} = 3 \cdot 6 + 5 \cdot (-1) + (-7) \cdot 0 = 13; \text{ Row 2} = 0 \cdot 6 + (-1) \cdot (-1) + 8 \cdot 0 = 1. BC = \begin{pmatrix} 13 \\ 1 \end{pmatrix}.$$

$$A(BC): \text{ Row 1} = 3 \cdot 13 + 0 \cdot 1 = 39; \text{ Row 2} = (-4) \cdot 13 + (-1) \cdot 1 = -53. ABC = \begin{pmatrix} 39 \\ -53 \end{pmatrix}, \text{ so } (ABC)^T = \begin{pmatrix} 39 & -53 \end{pmatrix}.$$

$$C^T B^T = \begin{pmatrix} 6 & -1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 5 & -1 \\ -7 & 8 \end{pmatrix} = [18 - 5 + 0, 0 + 1 + 0] = \begin{pmatrix} 13 & 1 \end{pmatrix}.$$

$$\text{Then } \begin{pmatrix} 13 & 1 \end{pmatrix} A^T = \begin{pmatrix} 13 & 1 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 0 & -1 \end{pmatrix} = [13 \cdot 3 + 1 \cdot 0, 13 \cdot (-4) + 1 \cdot (-1)] = \begin{pmatrix} 39 & -53 \end{pmatrix} \checkmark$$

Integral powers of a square matrix

Definition: For a square matrix A and positive integer n , $A^n = A \cdot A \cdot \dots \cdot A$ (n factors); so $A^2 = AA$ and $A^{p+q} = A^p A^q$.

Square matrix A and positive integer n . $A^n = A \cdot A \cdot \dots \cdot A$ (n factors); $A^2 = AA$, $A^{p+q} = A^p A^q$.

Only a square matrix can be multiplied by itself. $A^2 = AA$, $A^3 = A^2A$, $A^4 = A^3A = A^2A^2$, and in general $A^{p+q} = A^p A^q = A^q A^p$. Because $AB \neq BA$ in general, the school identity $(a + b)^2 = a^2 + 2ab + b^2$ needs care with matrices.

മലയാളത്തിൽ

Square matrix-നെ മാത്രമേ അതിനോടു തന്നെ ഗുണിക്കാൻ പറ്റൂ. A^8 കാണാൻ എട്ട് തവണ ഗുണിക്കേണ്ട: A^2 കണ്ടാൽ $A^4 = A^2 A^2$, പിന്നെ $A^8 = A^4 A^4$, മൂന്ന് step മതി. $(A + B)^2$ വിടർത്തുമ്പോൾ AB -യും BA -യും വേറെ വേറെ എഴുതണം, കാരണം അവ സാധാരണ ഒന്നല്ല.

$$A^2 = AA, A^3 = A^2 A = AA^2, A^{p+q} = A^p A^q$$

$$(A + B)^2 = A^2 + AB + BA + B^2$$

$$(A + B)^2 = A^2 + 2AB + B^2 \text{ if and only if } AB = BA$$

EXAMPLE

$$\text{If } A = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}, \text{ find } A^4.$$

$$A^2 = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}: \text{Row 1} \times \text{Col 1} = 1 + 8 = 9; \text{Row 1} \times \text{Col 2} = 2 + 6 = 8; \text{Row 2} \times \text{Col 1} = 4 + 12 = 16; \text{Row 2} \times \text{Col 2} = 8 + 9 = 17. A^2 = \begin{pmatrix} 9 & 8 \\ 16 & 17 \end{pmatrix}.$$

$$A^4 = A^2 A^2 = \begin{pmatrix} 9 & 8 \\ 16 & 17 \end{pmatrix} \begin{pmatrix} 9 & 8 \\ 16 & 17 \end{pmatrix}: \text{Row 1} \times \text{Col 1} = 81 + 128 = 209; \text{Row 1} \times \text{Col 2} = 72 + 136 = 208; \text{Row 2} \times \text{Col 1} = 144 + 272 = 416; \text{Row 2} \times \text{Col 2} = 128 + 289 = 417.$$

$$A^4 = \begin{pmatrix} 209 & 208 \\ 416 & 417 \end{pmatrix}$$

EXAMPLE

$$\text{If } A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}, \text{ find } A^8.$$

$$A^2 = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 2 & 1 \cdot 0 + 0 \cdot 3 \\ 2 \cdot 1 + 3 \cdot 2 & 2 \cdot 0 + 3 \cdot 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 8 & 9 \end{pmatrix}$$

$$A^4 = A^2 A^2 = \begin{pmatrix} 1 + 0 & 0 + 0 \\ 8 + 72 & 0 + 81 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 80 & 81 \end{pmatrix}$$

$$A^8 = A^4 A^4 = \begin{pmatrix} 1 + 0 & 0 + 0 \\ 80 + 81 \cdot 80 & 0 + 81 \cdot 81 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 6560 & 6561 \end{pmatrix}$$

Cofactors and the adjoint matrix

Cofactor: $A_{ij} = (-1)^{i+j}M_{ij}$, where the minor M_{ij} is the determinant left after deleting row i and column j .

$A_{ij} = (-1)^{i+j}M_{ij}$; ഇവിടെ minor M_{ij} എന്നത് row i -യും column j -യും ഒഴിവാക്കിയ ശേഷം ബാക്കിയുള്ള determinant ആണ്.

Adjoint: $\text{adj } A$ is the transpose of the matrix of cofactors of a square matrix A ; $A(\text{adj } A) = (\text{adj } A)A = |A|I$.

Square matrix A -യുടെ cofactors-ന്റെ matrix-ന്റെ transpose ആണ് $\text{adj } A$; $A(\text{adj } A) = (\text{adj } A)A = |A|I$.

For a square matrix A , the minor M_{ij} is the determinant left after deleting row i and column j . The cofactor is $A_{ij} = (-1)^{i+j}M_{ij}$. The **adjoint** (adjugate) of A , $\text{adj } A$, is the **transpose** of the matrix of cofactors. Its key property is $A(\text{adj } A) = (\text{adj } A)A = |A|I$.

മലയാളത്തിൽ

ഓരോ element-നും ഒരു cofactor ഉണ്ട്. ആ element-ന്റെ row-ഉം column-ഉം മറച്ചു വെച്ച് ബാക്കി determinant കണ്ട്, $(-1)^{i+j}$ എന്ന sign ചേർക്കുക. Sign pattern ചെസ്സ് ബോർഡ് പോലെ $+$ $-$ $+$ ആണ്. എല്ലാ cofactors-ഉം matrix ആയി എഴുതി transpose എടുത്താൽ $\text{adj } A$. Transpose മറക്കുന്നതാണ് exam-ൽ ഏറ്റവും സാധാരണ തെറ്റ്.

$$A_{ij} = (-1)^{i+j}M_{ij}; \text{ sign pattern } \begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$\text{adj } A = [A_{ij}]^T$$

$$A(\text{adj } A) = (\text{adj } A)A = |A| I$$

$$2 \times 2 \text{ shortcut: } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \text{adj } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \text{ (swap the diagonal, change sign of the other two)}$$

EXAMPLE

For $A = \begin{pmatrix} 2 & -3 \\ 4 & 5 \end{pmatrix}$, verify $A(\text{adj } A) = (\text{adj } A)A = |A|I_2$.

$A_{11} = 5, A_{12} = -4, A_{21} = -(-3) = 3, A_{22} = 2$. Cofactor matrix $\begin{pmatrix} 5 & -4 \\ 3 & 2 \end{pmatrix}$, so $\text{adj } A = \begin{pmatrix} 5 & 3 \\ -4 & 2 \end{pmatrix}$.

$$|A| = 2 \cdot 5 - (-3) \cdot 4 = 10 + 12 = 22.$$

$A(\text{adj } A)$: Row 1 $\times$ Col 1 = $2 \cdot 5 + (-3)(-4) = 22$; Row 1 $\times$ Col 2 = $2 \cdot 3 + (-3) \cdot 2 = 0$; Row 2 $\times$ Col 1 = $4 \cdot 5 + 5 \cdot (-4) = 0$; Row 2 $\times$ Col 2 = $4 \cdot 3 + 5 \cdot 2 = 22$. Result $\begin{pmatrix} 22 & 0 \\ 0 & 22 \end{pmatrix} = 22I_2$.

$(\text{adj } A)A$: Row 1 $\times$ Col 1 = $5 \cdot 2 + 3 \cdot 4 = 22$; Row 1 $\times$ Col 2 = $5 \cdot (-3) + 3 \cdot 5 = 0$; Row 2 $\times$ Col 1 = $(-4) \cdot 2 + 2 \cdot 4 = 0$; Row 2 $\times$ Col 2 = $(-4)(-3) + 2 \cdot 5 = 22$. Result $22I_2 = |A|I_2 \checkmark$

EXAMPLE

Find $\text{adj } A$ for $A = \begin{pmatrix} 1 & 2 & 3 \\ -4 & 5 & 0 \\ -1 & -2 & 1 \end{pmatrix}$ and verify $A(\text{adj } A) = |A|I$.

$$A_{11} = + \begin{vmatrix} 5 & 0 \\ -2 & 1 \end{vmatrix} = 5 - 0 = 5; A_{12} = - \begin{vmatrix} -4 & 0 \\ -1 & 1 \end{vmatrix} = -(-4 - 0) = 4; A_{13} = + \begin{vmatrix} -4 & 5 \\ -1 & -2 \end{vmatrix} = 8 + 5 = 13$$

$$A_{21} = - \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} = -(2 + 6) = -8; A_{22} = + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 1 + 3 = 4; A_{23} = - \begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} = -(-2 + 2) = 0$$

$$A_{31} = + \begin{vmatrix} 2 & 3 \\ 5 & 0 \end{vmatrix} = 0 - 15 = -15; A_{32} = - \begin{vmatrix} 1 & 3 \\ -4 & 0 \end{vmatrix} = -(0 + 12) = -12; A_{33} = + \begin{vmatrix} 1 & 2 \\ -4 & 5 \end{vmatrix} = 5 + 8 = 13$$

$$\text{adj } A = \begin{pmatrix} 5 & 4 & 13 \\ -8 & 4 & 0 \\ -15 & -12 & 13 \end{pmatrix}^T = \begin{pmatrix} 5 & -8 & -15 \\ 4 & 4 & -12 \\ 13 & 0 & 13 \end{pmatrix}$$

$$|A| \text{ (along column 3)} = 3 \cdot A_{13} + 0 \cdot A_{23} + 1 \cdot A_{33} = 39 + 0 + 13 = 52.$$

$A(\text{adj } A)$, row by column:

$$\text{Row 1 (1, 2, 3): } \cdot \text{Col 1 (5, 4, 13)} = 5 + 8 + 39 = 52; \cdot \text{Col 2 (-8, 4, 0)} = -8 + 8 + 0 = 0; \cdot \text{Col 3 (-15, -12, 13)} = -15 - 24 + 39 = 0$$

$$\text{Row 2 (-4, 5, 0): } \cdot \text{Col 1} = -20 + 20 + 0 = 0; \cdot \text{Col 2} = 32 + 20 + 0 = 52; \cdot \text{Col 3} = 60 - 60 + 0 = 0$$

$$\text{Row 3 (-1, -2, 1): } \cdot \text{Col 1} = -5 - 8 + 13 = 0; \cdot \text{Col 2} = 8 - 8 + 0 = 0; \cdot \text{Col 3} = 15 + 24 + 13 = 52$$

$$A(\text{adj } A) = \begin{pmatrix} 52 & 0 & 0 \\ 0 & 52 & 0 \\ 0 & 0 & 52 \end{pmatrix} = 52I = |A|I \checkmark \text{ (the product } (\text{adj } A)A \text{ also gives } 52I).$$

Exam tip: A quick check of your cofactors: row i of A times column i of $\text{adj } A$ must give $|A|$, and every off-diagonal product must give 0.

Trace of a matrix

Definition: The trace $\text{tr}(A)$ of a square matrix A is the sum of the elements on its principal diagonal: $a_{11} + a_{22} + \dots + a_{nn}$.

Square matrix A -യുടെ principal diagonal-ലെ elements-ന്റെ sum ആണ് trace $\text{tr}(A)$: $a_{11} + a_{22} + \dots + a_{nn}$.

The trace of a square matrix, $\text{tr}(A)$, is the sum of the elements on its principal diagonal. The trace is linear and does not change under transpose or under a cyclic change of order in a product. However, $\text{tr}(AB)$ is generally not $\text{tr}(A) \cdot \text{tr}(B)$.

മലയാളത്തിൽ

Trace എന്നാൽ diagonal-ലെ elements മാത്രം കൂട്ടിയ തുക. ബാക്കി elements-ന് ഇവിടെ ഒരു റോളമില്ല. $\text{tr}(AB) = \text{tr}(BA)$ ആണ്, AB -യും BA -യും വേറെ matrices ആണെങ്കിലും.

$$\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$$

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B); \text{tr}(kA) = k \cdot \text{tr}(A); \text{tr}(A^T) = \text{tr}(A)$$

$$\text{tr}(AB) = \text{tr}(BA); \text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$$

$$\text{In general } \text{tr}(AB) \neq \text{tr}(A) \cdot \text{tr}(B)$$

EXAMPLE

Find $\text{tr}(A)$ for $A = \begin{pmatrix} 2 & 3 & 5 \\ 6 & 8 & 4 \\ 9 & 1 & -3 \end{pmatrix}$, for $B = \begin{pmatrix} 8 & 7 \\ 5 & 6 \end{pmatrix}$, and for I_n .

$$\text{tr}(A) = 2 + 8 + (-3) = 7. \text{tr}(B) = 8 + 6 = 14.$$

$$\text{tr}(I_2) = 1 + 1 = 2, \text{tr}(I_3) = 3, \text{and } \text{tr}(I_n) = 1 + 1 + \dots + 1 \text{ (n times)} = n.$$

EXAMPLE

Using $A = \begin{pmatrix} 5 & 4 \\ 2 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 6 & 8 \\ 2 & 9 \end{pmatrix}$ from the multiplication topic, check $\text{tr}(AB) = \text{tr}(BA)$.

$$AB = \begin{pmatrix} 38 & 76 \\ 26 & 79 \end{pmatrix} \Rightarrow \text{tr}(AB) = 38 + 79 = 117. BA = \begin{pmatrix} 46 & 80 \\ 28 & 71 \end{pmatrix} \Rightarrow \text{tr}(BA) = 46 + 71 = 117 \checkmark$$

But $\text{tr}(A) \cdot \text{tr}(B) = 12 \times 15 = 180 \neq 117$.

Sub-matrices and minors

Sub-matrix: A matrix obtained from A by deleting some of its rows and/or columns.

A-യിൽ നിന്ന് ചില rows-ഉം/അല്ലെങ്കിൽ columns-ഉം ഒഴിവാക്കി കിട്ടുന്ന matrix.

Minor of order r: The determinant of an $r \times r$ square sub-matrix of A.

A-യുടെ ഒരു $r \times r$ square sub-matrix-ന്റെ determinant.

A **sub-matrix** of A is any matrix obtained by deleting some rows and/or some columns of A. A **minor of order r** of an $m \times n$ matrix A is the determinant of an $r \times r$ square sub-matrix, i.e. obtained by deleting $(m - r)$ rows and $(n - r)$ columns. Minors are what we use to define cofactors and rank.

മലയാളത്തിൽ

ഒരു matrix-ൽ നിന്ന് ചില rows-ഉം columns-ഉം വെട്ടിമാറ്റിയാൽ കിട്ടുന്നത് sub-matrix. ആ sub-matrix square ആണെങ്കിൽ, അതിന്റെ determinant ആണ് minor. അതായത് sub-matrix ഒരു matrix, minor ഒരു number.

Minor of order r = determinant of an $r \times r$ sub-matrix

EXAMPLE

List the order-2 minors of $A = \begin{pmatrix} a & h & g \\ h & b & f \\ g & f & c \end{pmatrix}$ that use rows 1, 2.

Delete row 3 and one column each time:

delete column 3: $\begin{vmatrix} a & h \\ h & b \end{vmatrix} = ab - h^2$; delete column 2: $\begin{vmatrix} a & g \\ h & f \end{vmatrix} = af - gh$; delete column 1: $\begin{vmatrix} h & g \\ b & f \end{vmatrix} = hf - bg$.

Also $\begin{pmatrix} a & h \\ h & b \end{pmatrix}$ (rows 1-2, columns 1-2) is a 2×2 sub-matrix of A .

Exam tip: The unit writes the minor in square brackets; strictly, a minor is the determinant (a number), e.g. $ab - h^2$, not the sub-matrix itself.

Inverse of a matrix

Inverse: For a square matrix A of order n , the matrix A^{-1} with $AA^{-1} = A^{-1}A = I_n$; $A^{-1} = (1/|A|) \text{adj } A$.

Order n square matrix A —യ്ക്കു് $AA^{-1} = A^{-1}A = I_n$ ആകുന്ന matrix A^{-1} ആണ് inverse; $A^{-1} = (1/|A|) \text{adj } A$.

Singular / non-singular: A square matrix is singular if $|A| = 0$ (no inverse) and non-singular if $|A| \neq 0$ (inverse exists).

$|A| = 0$ ആണെങ്കിൽ square matrix singular (inverse ഇല്ല); $|A| \neq 0$ ആണെങ്കിൽ non-singular (inverse ഉണ്ട്).

For a square matrix A of order n , if there is a matrix B with $AB = BA = I_n$, then B is the inverse A^{-1} . It plays the role of the reciprocal $1/a$ for numbers. A^{-1} exists if and only if $|A| \neq 0$ (A is **non-singular**); if $|A| = 0$, A is **singular** and has no inverse. Dividing $A(\text{adj } A) = |A|I$ by $|A|$ gives the formula below.

മലയാളത്തിൽ

$5 \times (1/5) = 1$ എന്ന പോലെ, $A \times A^{-1} = I$. Inverse കാണാൻ മൂന്ന് step: ആദ്യം $|A|$ കാണുക, അത് പൂജ്യമാണെങ്കിൽ inverse ഇല്ല, അവിടെ നിർത്താം. പൂജ്യമല്ലെങ്കിൽ $\text{adj } A$ കണ്ടു് $|A|$ കൊണ്ട് ഹരിക്കുക. $(AB)^{-1} = B^{-1}A^{-1}$ എന്ന reversal law ഓർക്കുക, transpose-ൽ കണ്ട socks-shoes rule തന്നെ.

$$A^{-1} = (1/|A|) \operatorname{adj} A, \text{ provided } |A| \neq 0$$

$$2 \times 2: A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = (1/(ad - bc)) \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\text{Inverse is unique; } (A^{-1})^{-1} = A$$

$$(AB)^{-1} = B^{-1}A^{-1} \text{ (reversal law)}$$

$$(A^{-1})^T = (A^T)^{-1}$$

EXAMPLE

$$\text{Find } A^{-1} \text{ where } A = \begin{pmatrix} 4 & 2 & 3 \\ 0 & -1 & -2 \\ 5 & -3 & 6 \end{pmatrix}.$$

$$|A| = 4[(-1)(6) - (-2)(-3)] - 2[(0)(6) - (-2)(5)] + 3[(0)(-3) - (-1)(5)] = 4(-12) - 2(10) + 3(5) = -48 - 20 + 15 = -53 \neq 0, \text{ so } A^{-1} \text{ exists.}$$

$$A_{11} = (-6 - 6) = -12; A_{12} = -(0 + 10) = -10; A_{13} = (0 + 5) = 5$$

$$A_{21} = -(12 + 9) = -21; A_{22} = (24 - 15) = 9; A_{23} = -(-12 - 10) = 22$$

$$A_{31} = (-4 + 3) = -1; A_{32} = -(-8 - 0) = 8; A_{33} = (-4 - 0) = -4$$

$$\operatorname{adj} A = \begin{pmatrix} -12 & -10 & 5 \\ -21 & 9 & 22 \\ -1 & 8 & -4 \end{pmatrix}^T = \begin{pmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{pmatrix}$$

$$A^{-1} = (1/-53) \begin{pmatrix} -12 & -21 & -1 \\ -10 & 9 & 8 \\ 5 & 22 & -4 \end{pmatrix} = (1/53) \begin{pmatrix} 12 & 21 & 1 \\ 10 & -9 & -8 \\ -5 & -22 & 4 \end{pmatrix}$$

$$\text{Check one entry: Row 1 of } A \times \text{Col 1 of } \operatorname{adj} A = 4(-12) + 2(-10) + 3(5) = -53 = |A| \checkmark$$

EXAMPLE

Find the inverse of $A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{pmatrix}$.

$$|A| = 1[(-1)(-1) - 3 \cdot 2] - 1[2(-1) - 3 \cdot 3] + 1[2 \cdot 2 - (-1) \cdot 3] = -5 + 11 + 7 = 13 \neq 0.$$

$$A_{11} = (1 - 6) = -5; A_{12} = -(-2 - 9) = 11; A_{13} = (4 + 3) = 7$$

$$A_{21} = -(-1 - 2) = 3; A_{22} = (-1 - 3) = -4; A_{23} = -(2 - 3) = 1$$

$$A_{31} = (3 + 1) = 4; A_{32} = -(3 - 2) = -1; A_{33} = (-1 - 2) = -3$$

$$\text{Cofactor matrix} = \begin{pmatrix} -5 & 11 & 7 \\ 3 & -4 & 1 \\ 4 & -1 & -3 \end{pmatrix}, \text{adj } A = \begin{pmatrix} -5 & 3 & 4 \\ 11 & -4 & -1 \\ 7 & 1 & -3 \end{pmatrix}$$

$$A^{-1} = (1/13) \begin{pmatrix} -5 & 3 & 4 \\ 11 & -4 & -1 \\ 7 & 1 & -3 \end{pmatrix}$$

$$\text{Check: Row 2 of } A \times \text{Col 3 of adj } A = 2 \cdot 4 + (-1)(-1) + 3(-3) = 8 + 1 - 9 = 0 \checkmark$$

Exam tip: In this example (Example 10) the unit's cofactor matrix has $A_{32} = -2$ and its adj A has $+1$ in row 2, column 3. Both are misprints: $A_{32} = -(1 \cdot 3 - 1 \cdot 2) = -1$, so the $(2, 3)$ entry of adj A is -1 . The unit's final A^{-1} is correct.

EXAMPLE

For $A = \begin{pmatrix} 2 & 3 \\ -1 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 4 \\ 5 & -2 \end{pmatrix}$, verify $(AB)^{-1} = B^{-1}A^{-1}$.

$$|A| = 8 + 3 = 11, \text{adj } A = \begin{pmatrix} 4 & -3 \\ 1 & 2 \end{pmatrix}, A^{-1} = (1/11) \begin{pmatrix} 4 & -3 \\ 1 & 2 \end{pmatrix}.$$

$$|B| = -2 - 20 = -22, \text{adj } B = \begin{pmatrix} -2 & -4 \\ -5 & 1 \end{pmatrix}, B^{-1} = (-1/22) \begin{pmatrix} -2 & -4 \\ -5 & 1 \end{pmatrix}.$$

$$AB: \text{Row } 1 \times \text{Col } 1 = 2 + 15 = 17; \text{Row } 1 \times \text{Col } 2 = 8 - 6 = 2; \text{Row } 2 \times \text{Col } 1 = -1 + 20 = 19; \text{Row } 2 \times \text{Col } 2 = -4 - 8 = -12. AB = \begin{pmatrix} 17 & 2 \\ 19 & -12 \end{pmatrix}.$$

$$|AB| = -204 - 38 = -242, \text{adj}(AB) = \begin{pmatrix} -12 & -2 \\ -19 & 17 \end{pmatrix}, \text{so } (AB)^{-1} = (-1/242) \begin{pmatrix} -12 & -2 \\ -19 & 17 \end{pmatrix} \dots (1)$$

$$B^{-1}A^{-1} = (-1/242) \begin{pmatrix} -2 & -4 \\ -5 & 1 \end{pmatrix} \begin{pmatrix} 4 & -3 \\ 1 & 2 \end{pmatrix}. \text{Row } 1 \times \text{Col } 1 = -8 - 4 = -12; \text{Row } 1 \times \text{Col } 2 = 6 - 8 = -2; \text{Row } 2 \times \text{Col } 1 = -20 + 1 = -19; \text{Row } 2 \times \text{Col } 2 = 15 + 2 = 17.$$

$$B^{-1}A^{-1} = (-1/242) \begin{pmatrix} -12 & -2 \\ -19 & 17 \end{pmatrix}, \text{which equals (1) } \checkmark$$

Orthogonal matrix

Definition: A square matrix A is orthogonal if $A^T A = AA^T = I$, i.e. $A^{-1} = A^T$.

$A^T A = AA^T = I$, അതായത് $A^{-1} = A^T$ ആണെങ്കിൽ square matrix A orthogonal ആണ്.

A square matrix A is orthogonal if $A^T A = I$ (equivalently $AA^T = I$, so $A^{-1} = A^T$). Taking determinants, $|A^T||A| = |A|^2 = 1$, so $|A| = \pm 1$ and an orthogonal matrix is always non-singular. Identity matrices are orthogonal, and products, transposes and inverses of orthogonal matrices are orthogonal.

മലയാളത്തിൽ

Transpose തന്നെ inverse ആയി വരുന്ന matrix ആണ് orthogonal. Inverse കാണാൻ adjoint ഒന്നും വേണ്ട, transpose എടുത്താൽ മതി. ഇത്തരം matrices-ന്റെ determinant 1 അല്ലെങ്കിൽ -1 മാത്രം. Rotation പോലുള്ള geometry-യിലെ transformations ഇങ്ങനെയാണ്.

$$A^T A = A A^T = I \Leftrightarrow A^{-1} = A^T$$

$$|A| = \pm 1$$

EXAMPLE

Show that $A = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}$ is orthogonal.

$$A^T = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix}.$$

$A^T A$: Row 1 $\times$ Col 1 = $9/25 + 16/25 = 1$; Row 1 $\times$ Col 2 = $-12/25 + 12/25 = 0$; Row 2 $\times$ Col 1 = $-12/25 + 12/25 = 0$; Row 2 $\times$ Col 2 = $16/25 + 9/25 = 1$.

$A^T A = I$, so A is orthogonal. Also $|A| = 9/25 + 16/25 = 1$.

Symmetric and skew-symmetric matrices

Symmetric matrix: A square matrix with $A^T = A$, i.e. $a_{ij} = a_{ji}$ for all i, j .

$A^T = A$, അതായത് എല്ലാ i, j -നും $a_{ij} = a_{ji}$ ആയ square matrix.

Skew-symmetric matrix: A square matrix with $A^T = -A$, i.e. $a_{ij} = -a_{ji}$; all its diagonal elements are 0.

$A^T = -A$, അതായത് $a_{ij} = -a_{ji}$ ആയ square matrix; അതിന്റെ diagonal elements എല്ലാം 0 ആണ്.

A square matrix is **symmetric** if $A^T = A$, i.e. $a_{ij} = a_{ji}$ (mirror image across the diagonal). It is **skew-symmetric** if $A^T = -A$, i.e. $a_{ij} = -a_{ji}$; then $a_{ii} = -a_{ii}$, so every diagonal element is 0. For any square A , $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric, so every square matrix splits uniquely into a symmetric part plus a skew-symmetric part.

മലയാളത്തിൽ

Diagonal-നെ ഒരു കണ്ണാടി ആയി കരുതുക. രണ്ടു വശത്തും ഒരേ numbers ആണെങ്കിൽ symmetric. ഒരേ numbers, പക്ഷേ sign മാറിയിട്ടാണെങ്കിൽ skew-symmetric, അപ്പോൾ diagonal മുഴുവൻ പൂജ്യമായിരിക്കും. ഏത് square matrix-നെയും P (symmetric) + Q (skew-symmetric) ആയി പിരിച്ചെഴുതാം.

Symmetric: $A^T = A$, e.g. $\begin{pmatrix} a & b & c \\ b & e & d \\ c & d & f \end{pmatrix}$

Skew-symmetric: $A^T = -A$, e.g. $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$

AA^T is symmetric, since $(AA^T)^T = (A^T)^T A^T = AA^T$

If A, B symmetric and $AB = BA$, then AB is symmetric

$A = P + Q$, $P = \frac{1}{2}(A + A^T)$ (symmetric), $Q = \frac{1}{2}(A - A^T)$ (skew-symmetric)

EXAMPLE

If $A = \begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix}$, show that $P = \frac{1}{2}(A + A^T)$ is symmetric, $Q = \frac{1}{2}(A - A^T)$ is skew-symmetric, and $A = P + Q$.

$$A^T = \begin{pmatrix} 3 & -2 \\ 5 & 4 \end{pmatrix}.$$

$$P = \frac{1}{2} \begin{pmatrix} 3+3 & 5-2 \\ -2+5 & 4+4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 6 & 3 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 3 & 3/2 \\ 3/2 & 4 \end{pmatrix}. P^T = \begin{pmatrix} 3 & 3/2 \\ 3/2 & 4 \end{pmatrix} = P, \text{ so } P \text{ is symmetric.}$$

$$Q = \frac{1}{2} \begin{pmatrix} 3-3 & 5+2 \\ -2-5 & 4-4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 7 \\ -7 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 7/2 \\ -7/2 & 0 \end{pmatrix}. Q^T = \begin{pmatrix} 0 & -7/2 \\ 7/2 & 0 \end{pmatrix} = -Q, \text{ so } Q \text{ is skew-symmetric.}$$

$$P + Q = \begin{pmatrix} 3+0 & 3/2+7/2 \\ 3/2-7/2 & 4+0 \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ -2 & 4 \end{pmatrix} = A \quad \checkmark$$

Idempotent matrix

Definition: A square matrix A is idempotent if $A^2 = A$.

$A^2 = A$ ആണെങ്കിൽ square matrix A idempotent ആണ്.

A square matrix A is idempotent if $A^2 = A$: multiplying it by itself gives the same matrix back. Such a matrix acts like a projection: applying it twice has the same effect as applying it once. Idempotent matrices appear in linear algebra, geometry, computer graphics and network analysis.

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$A^2 = A$ ആയാൽ idempotent. ഒരിക്കൽ ചെയ്താലും രണ്ടു തവണ ചെയ്താലും ഫലം ഒന്നു തന്നെ: light switch "ON" അമർത്തിയാൽ on ആകും, വീണ്ടും "ON" അമർത്തിയാലും on തന്നെ. I-ഉം O-ഉം ഇതിന്റെ എളുപ്പമുള്ള ഉദാഹരണങ്ങളാണ്.

A is idempotent $\Leftrightarrow A^2 = A$ (then $A^3 = A^2A = A \cdot A = A$, and so on)

EXAMPLE

Show that $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ is idempotent.

$A^2 = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$: Row 1 $\times$ Col 1 = $4 - 2 = 2$; Row 1 $\times$ Col 2 = $-4 + 2 = -2$; Row 2 $\times$ Col 1 = $2 - 1 = 1$;
Row 2 $\times$ Col 2 = $-2 + 1 = -1$.

$A^2 = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} = A$, so A is idempotent.

Partitioned matrices

Definition: A partitioned matrix is a matrix divided by horizontal and vertical lines into sub-matrices (blocks), which can be treated like single elements in matrix operations.

Horizontal, vertical lines ഉപയോഗിച്ച് sub-matrices (blocks) ആയി വിഭജിച്ച matrix ആണ് partitioned matrix; matrix operations-ൽ ഈ blocks-നെ single elements പോലെ കൈകാര്യം ചെയ്യാം.

A matrix can be cut by horizontal and vertical lines into smaller blocks (sub-matrices). If two matrices are partitioned conformably, the blocks can be added, subtracted or multiplied as if they were single numbers. The main computational benefit is in multiplication of large matrices.

മലയാളത്തിൽ

വലിയ matrix-നെ ചെറിയ കഷണങ്ങളാക്കി (blocks) ഓരോ block-നെയും ഒരു number പോലെ കൈകാര്യം ചെയ്യുന്നതാണ് partition. ഒരു വലിയ പഠനത്തെ plots ആക്കി തിരിച്ച് ഓരോന്നും വെവ്വേറെ കണക്കാക്കുന്നത് പോലെ. Multiply ചെയ്യുമ്പോൾ blocks-ന്റെ orders ചേരണം എന്നു മാത്രം.

$$A = (A_1 : A_2), B = (B_1 : B_2) \text{ same split} \Rightarrow A \pm B = (A_1 \pm B_1 : A_2 \pm B_2)$$

$$A = (A_1 : A_2) \text{ with } A_1: m \times n_1, A_2: m \times n_2; B = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} \text{ with } B_1: n_1 \times p, B_2: n_2 \times p \Rightarrow AB = A_1B_1 + A_2B_2$$

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

EXAMPLE

$$\text{Let } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \text{ be split after column 2 as } A_1 = \begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}, A_2 = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \text{ and } B = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \text{ split after row 2 as } B_1 =$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, B_2 = \begin{pmatrix} 2 \end{pmatrix}. \text{ Find } AB \text{ by blocks.}$$

$$A_1B_1: \text{Row 1} = 1 \cdot 1 + 2 \cdot 0 = 1; \text{Row 2} = 4 \cdot 1 + 5 \cdot 0 = 4 \Rightarrow \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

$$A_2B_2 = \begin{pmatrix} 3 \cdot 2 \\ 6 \cdot 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}.$$

$$AB = A_1B_1 + A_2B_2 = \begin{pmatrix} 1 \\ 4 \\ 6 \\ 12 \end{pmatrix}. \text{ Directly: Row 1} = 1 + 0 + 6 = 7, \text{ Row 2} = 4 + 0 + 12 = 16 \checkmark$$

Rank of a matrix

Definition: The rank of a matrix A is the order r of its largest non-zero minor: some $r \times r$ minor is non-zero and every $(r + 1) \times (r + 1)$ minor is zero.

A -യിലെ ഏറ്റവും വലിയ non-zero minor-ന്റെ order r ആണ് rank: ഏതെങ്കിലും ഒരു $r \times r$ minor non-zero ആയിരിക്കും, എല്ലാ $(r + 1) \times (r + 1)$ minors-ഉം zero ആയിരിക്കും.

The rank r of a matrix A is the order of the largest non-zero minor in A : there is at least one $r \times r$ sub-matrix with non-zero determinant, and every $(r + 1) \times (r + 1)$ sub-matrix has determinant 0. The rank cannot exceed the smaller of the number of rows and columns; the null matrix has rank 0; a non-zero row or column matrix has rank 1; and $\text{rank}(A^T) = \text{rank}(A)$. Elementary row/column operations do not change the rank, so they can be used to simplify the matrix first.

മലയാളത്തിൽ

Rank കാണാൻ ഏറ്റവും വലിയ square sub-matrix-ൽ നിന്ന് തുടങ്ങുക. അതിന്റെ determinant പൂജ്യമല്ലെങ്കിൽ rank അതിന്റെ order. എല്ലാം പൂജ്യമാണെങ്കിൽ ഒരു size താഴോട്ട് ഇറങ്ങി നോക്കുക. ഒരു row മറ്റൊന്നിന്റെ multiple ആണെങ്കിൽ, ആ row പുതിയ വിവരമൊന്നും തരുന്നില്ല; rank കുറയും.

$$\text{rank}(A) = r \Leftrightarrow \text{some } r \times r \text{ minor} \neq 0 \text{ and every } (r+1) \times (r+1) \text{ minor} = 0$$

$$\text{rank}(A_{m \times n}) \leq \min(m, n); \text{rank}(O) = 0; \text{rank}(A^T) = \text{rank}(A)$$

EXAMPLE

Find the rank of (i) $\begin{pmatrix} 5 & 2 \\ -2 & -3 \end{pmatrix}$ (ii) $\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}$ (iii) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

$$(i) \begin{vmatrix} 5 & 2 \\ -2 & -3 \end{vmatrix} = -15 + 4 = -11 \neq 0, \text{ so rank} = 2.$$

$$(ii) \begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix} = 12 - 12 = 0, \text{ but a } 1 \times 1 \text{ minor such as } 3 \neq 0, \text{ so rank} = 1. (\text{Row 2 is } 2 \times \text{row 1.})$$

(iii) Every minor is 0, so rank = 0.

EXAMPLE

Find the rank of $A = \begin{pmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{pmatrix}$.

A is 2×5 , so rank ≤ 2 . Take columns 1 and 2: $\begin{vmatrix} 2 & -1 \\ 3 & 2 \end{vmatrix} = 4 + 3 = 7 \neq 0$.

A non-zero 2×2 minor exists, so rank(A) = 2.

EXAMPLE

Find the rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$.

$$|A| = 1(4 - 0) - 2(2 - 6) + 3(0 - 4) = 4 + 8 - 12 = 0, \text{ so rank} < 3.$$

$$\text{Take rows 1, 3 and columns 1, 2: } \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 0 - 2 = -2 \neq 0. \text{ So rank}(A) = 2.$$

Exam tip: One non-zero minor is enough to prove the rank is at least that order; you do not need "any" minor to be non-zero. The unit also reduces Example 13 by column operations; that working is garbled in the text, and the short minor argument above gives the same answer, rank 2. Row/column operations preserve the rank, not the value of every determinant.

Application: solving linear equations by the matrix method

Matrix method: Write the system as $AX = B$; if $|A| \neq 0$, its unique solution is $X = A^{-1}B$.

System-നെ $AX = B$ ആയി എഴുതുക; $|A| \neq 0$ ആണെങ്കിൽ അതിന്റെ unique solution $X = A^{-1}B$ ആണ്.

Consistent system: A system with at least one solution; a system with no solution is inconsistent.

കറഞ്ഞത് ഒരു solution എങ്കിലും ഉള്ള system consistent ആണ്; solution ഇല്ലാത്ത system inconsistent ആണ്.

Homogeneous system: A system $AX = O$ (all constants zero); it is always consistent.

എല്ലാ constants-ഉം zero ആയ $AX = O$ system; അത് എപ്പോഴും consistent ആണ്.

A system of n linear equations in n unknowns can be written as $AX = B$, where A is the coefficient matrix, X the column of unknowns and B the column of constants. If $|A| \neq 0$, the system has a unique solution $X = A^{-1}B$. If $|A| = 0$, find $(\text{adj } A)B$: if it is not O , the system has no solution (inconsistent); if it is O , the system has infinitely many solutions or no solution, and you check by solving. A system with a solution is called consistent. If $B = O$ the system is homogeneous and is always consistent ($x = y = z = 0$ works).

മലയാളത്തിൽ

Equations-നെ $AX = B$ എന്ന ഒരു matrix equation ആക്കി മാറ്റുക. $|A| \neq 0$ ആണെങ്കിൽ $X = A^{-1}B$, ഒരു ഉത്തരം. $|A| = 0$ ആണെങ്കിൽ $(\text{adj } A)B$ കാണുക: അത് പൂജ്യമല്ലെങ്കിൽ ഉത്തരമില്ല; പൂജ്യമാണെങ്കിൽ ഒരു variable-നെ k ആയി എടുത്ത് solve ചെയ്യുക, പിന്നെ ബാക്കിയുള്ള equation-ൽ substitute ചെയ്ത് ഉറപ്പിക്കുക; ശരിയായാൽ infinite solutions, ഇല്ലെങ്കിൽ solution ഇല്ല. കടയിലെ bill-ൽ നിന്ന് ഓരോ സാധനത്തിന്റെയും വില കണ്ടുപിടിക്കുന്ന കണക്കുകൾ ഇങ്ങനെ ചെയ്യാം.

$AX = B \Rightarrow X = A^{-1}B = (1/|A|)(\text{adj } A)B$, if $|A| \neq 0$ (unique solution)

$|A| = 0$ and $(\text{adj } A)B \neq 0 \Rightarrow$ no solution

$|A| = 0$ and $(\text{adj } A)B = 0 \Rightarrow$ infinitely many solutions (or none); put one variable = k

EXAMPLE

2 pens, 3 notebooks and 1 book cost Rs 90; 1 pen, 4 notebooks and 2 books cost Rs 120; 2 pens, 4 notebooks and 5 books cost Rs 205. Find the cost of each by the matrix method.

Let the costs be x, y, z. Then $2x + 3y + z = 90$, $x + 4y + 2z = 120$, $2x + 4y + 5z = 205$, i.e.

$$\begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 2 & 4 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 90 \\ 120 \\ 205 \end{pmatrix}, AX = B.$$

$|A| = 2(20 - 8) - 3(5 - 4) + 1(4 - 8) = 24 - 3 - 4 = 17 \neq 0$, so a unique solution exists.

$A_{11} = 20 - 8 = 12$, $A_{12} = -(5 - 4) = -1$, $A_{13} = 4 - 8 = -4$; $A_{21} = -(15 - 4) = -11$, $A_{22} = 10 - 2 = 8$, $A_{23} = -(8 - 6) = -2$;
 $A_{31} = 6 - 4 = 2$, $A_{32} = -(4 - 1) = -3$, $A_{33} = 8 - 3 = 5$.

$$\text{adj } A = \begin{pmatrix} 12 & -11 & 2 \\ -1 & 8 & -3 \\ -4 & -2 & 5 \end{pmatrix}$$

$X = (1/17)(\text{adj } A)B$. Row by column with $B = (90, 120, 205)$:

Row 1: $12 \cdot 90 - 11 \cdot 120 + 2 \cdot 205 = 1080 - 1320 + 410 = 170$

Row 2: $-1 \cdot 90 + 8 \cdot 120 - 3 \cdot 205 = -90 + 960 - 615 = 255$

Row 3: $-4 \cdot 90 - 2 \cdot 120 + 5 \cdot 205 = -360 - 240 + 1025 = 425$

$$X = (1/17) \begin{pmatrix} 170 \\ 255 \\ 425 \end{pmatrix} = \begin{pmatrix} 10 \\ 15 \\ 25 \end{pmatrix}. \text{ A pen costs Rs 10, a notebook Rs 15, a book Rs 25.}$$

Check equation 1: $2(10) + 3(15) + 25 = 90 \checkmark$

EXAMPLE

Solve $x + y = 3$, $4x - 3y = 5$ by the matrix method.

$$A = \begin{pmatrix} 1 & 1 \\ 4 & -3 \end{pmatrix}, B = \begin{pmatrix} 3 \\ 5 \end{pmatrix}. |A| = -3 - 4 = -7 \neq 0.$$

$$\text{adj } A = \begin{pmatrix} -3 & -1 \\ -4 & 1 \end{pmatrix}, A^{-1} = (-1/7) \begin{pmatrix} -3 & -1 \\ -4 & 1 \end{pmatrix}.$$

$$X = (-1/7) \begin{pmatrix} -3 \cdot 3 - 1 \cdot 5 \\ -4 \cdot 3 + 1 \cdot 5 \end{pmatrix} = (-1/7) \begin{pmatrix} -14 \\ -7 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \text{ So } x = 2, y = 1.$$

EXAMPLE

Solve (i) $4x + 2y = 6$, $6x + 3y = 8$ and (ii) $2x - 3y = 3$, $6x - 9y = 5$.

$$(i) A = \begin{pmatrix} 4 & 2 \\ 6 & 3 \end{pmatrix}, |A| = 12 - 12 = 0. \text{ adj } A = \begin{pmatrix} 3 & -2 \\ -6 & 4 \end{pmatrix}. (\text{adj } A)B = \begin{pmatrix} 3 \cdot 6 - 2 \cdot 8 \\ -6 \cdot 6 + 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} \neq O. \text{ No solution (inconsistent).}$$

$$(ii) A = \begin{pmatrix} 2 & -3 \\ 6 & -9 \end{pmatrix}, |A| = -18 + 18 = 0. \text{ adj } A = \begin{pmatrix} -9 & 3 \\ -6 & 2 \end{pmatrix}. (\text{adj } A)B = \begin{pmatrix} -27 + 15 \\ -18 + 10 \end{pmatrix} = \begin{pmatrix} -12 \\ -8 \end{pmatrix} \neq O. \text{ No solution.}$$

EXAMPLE

Solve $3x + 6y - 4z = 3$, $3x - z = 0$, $12x - 6y - z = -3$.

$$A = \begin{pmatrix} 3 & 6 & -4 \\ 3 & 0 & -1 \\ 12 & -6 & -1 \end{pmatrix}. |A| = 3(0 - 6) - 6(-3 + 12) - 4(-18 - 0) = -18 - 54 + 72 = 0.$$

Cofactors: $A_{11} = -6$, $A_{12} = -9$, $A_{13} = -18$; $A_{21} = 30$, $A_{22} = 45$, $A_{23} = 90$; $A_{31} = -6$, $A_{32} = -9$, $A_{33} = -18$.

$$\text{adj } A = \begin{pmatrix} -6 & 30 & -6 \\ -9 & 45 & -9 \\ -18 & 90 & -18 \end{pmatrix}. (\text{adj } A)B \text{ with } B = (3, 0, -3): \text{Row 1} = -18 + 0 + 18 = 0; \text{Row 2} = -27 + 0 + 27 = 0;$$

Row 3 = $-54 + 0 + 54 = 0$. So $(\text{adj } A)B = O$.

Put $z = k$. From $3x - z = 0$: $x = k/3$. From the first equation: $k + 6y - 4k = 3 \Rightarrow 6y = 3 + 3k \Rightarrow y = (k + 1)/2$.

Check the third equation: $12(k/3) - 6(k + 1)/2 - k = 4k - 3k - 3 - k = -3 \checkmark$. So there are infinitely many solutions $x = k/3$, $y = (k + 1)/2$, $z = k$, k any real number.

Exam tip: The unit's answer key for this last system first derives $y = (k + 1)/2$ but then prints $y = (k + 1)/3$ in the final line; the correct answer is $y = (k + 1)/2$, as the check above shows. Also, $(\text{adj } A)B = O$ alone does not guarantee solutions (the unit's flow chart says "or no solution"), so always substitute back.

Quick recap

- A matrix of order $m \times n$ has m rows, n columns; a_{ij} sits in row i , column j .
- Row ($1 \times n$), column ($m \times 1$), square ($m = n$), rectangular ($m \neq n$), null (all zeros).
- Identity $\subset$ scalar $\subset$ diagonal; upper triangular has zeros below the diagonal, lower triangular has zeros above it.
- Equal matrices: same order and equal corresponding elements; compare positions to find unknowns.
- Add/subtract only same-order matrices, element by element; kA multiplies every element by k .
- AB needs columns of $A =$ rows of B ; $(m \times n)(n \times p) = m \times p$; each entry is row $\times$ column.

- $AB \neq BA$ in general; multiplication is associative and distributive; $IA = AI = A$.
- $(A^T)^T = A$, $(AB)^T = B^T A^T$; $(AB)^{-1} = B^{-1} A^{-1}$.
- $\text{adj } A$ = transpose of the cofactor matrix; $A(\text{adj } A) = (\text{adj } A)A = |A|I$.
- $A^{-1} = (1/|A|) \text{adj } A$ exists only when $|A| \neq 0$ (non-singular).
- $\text{tr}(A)$ = sum of diagonal elements; $\text{tr}(AB) = \text{tr}(BA)$.
- Orthogonal: $A^T A = I$, $|A| = \pm 1$. Symmetric: $A^T = A$. Skew-symmetric: $A^T = -A$ (zero diagonal). Idempotent: $A^2 = A$.
- Every square $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ (symmetric + skew-symmetric).
- Rank = order of the largest non-zero minor; $\text{rank} \leq \min(m, n)$.
- $AX = B$: $|A| \neq 0 \Rightarrow X = A^{-1}B$; $|A| = 0$ and $(\text{adj } A)B \neq O \Rightarrow$ no solution.

Check yourself

Q. 1. Write all possible orders of a matrix having 6 elements.

$m \cdot n = 6$ gives $1 \times 6, 2 \times 3, 3 \times 2$ and 6×1 .

Q. 2. Construct $A = [a_{ij}]_{2 \times 3}$ where $a_{ij} = i + 2j$.

$a_{11} = 3, a_{12} = 5, a_{13} = 7, a_{21} = 4, a_{22} = 6, a_{23} = 8$.

$$A = \begin{pmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{pmatrix}$$

Q. 3. Find x and y if $\begin{pmatrix} x+y & 2 \\ 5 & x-y \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 5 & 2 \end{pmatrix}$.

$x + y = 6$ and $x - y = 2$. Adding: $2x = 8 \Rightarrow x = 4$, then $y = 2$.

Q. 4. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$. Find AB and BA . Are they equal? Compare $\text{tr}(AB)$ and $\text{tr}(BA)$.

AB : Row 1 $\times$ Col 1 = $1 \cdot 2 + 2 \cdot 1 = 4$; Row 1 $\times$ Col 2 = $1 \cdot 0 + 2 \cdot 3 = 6$; Row 2 $\times$ Col 1 = $3 \cdot 2 + 4 \cdot 1 = 10$; Row 2 $\times$ Col 2 = $3 \cdot 0 + 4 \cdot 3 = 12$. $AB = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}$.

BA : Row 1 $\times$ Col 1 = $2 \cdot 1 + 0 \cdot 3 = 2$; Row 1 $\times$ Col 2 = $2 \cdot 2 + 0 \cdot 4 = 4$; Row 2 $\times$ Col 1 = $1 \cdot 1 + 3 \cdot 3 = 10$; Row 2 $\times$ Col 2 = $1 \cdot 2 + 3 \cdot 4 = 14$. $BA = \begin{pmatrix} 2 & 4 \\ 10 & 14 \end{pmatrix}$.

$AB \neq BA$, but $\text{tr}(AB) = 4 + 12 = 16 = 2 + 14 = \text{tr}(BA)$.

Q. 5. Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ and verify.

$|A| = 6 - 5 = 1 \neq 0$. $\text{adj } A = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$, so $A^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}$.

Check AA^{-1} : Row 1 $\times$ Col 1 = $6 - 5 = 1$; Row 1 $\times$ Col 2 = $-3 + 3 = 0$; Row 2 $\times$ Col 1 = $10 - 10 = 0$; Row 2 $\times$ Col 2 = $-5 + 6 = 1$. $AA^{-1} = I \checkmark$

Q. 6. Express $A = \begin{pmatrix} 2 & 6 \\ 0 & 4 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

$$A^T = \begin{pmatrix} 2 & 0 \\ 6 & 4 \end{pmatrix}.$$

$$P = \frac{1}{2}(A + A^T) = \frac{1}{2} \begin{pmatrix} 4 & 6 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix} \text{ (symmetric).}$$

$$Q = \frac{1}{2}(A - A^T) = \frac{1}{2} \begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} \text{ (skew-symmetric).}$$

$$P + Q = \begin{pmatrix} 2 & 6 \\ 0 & 4 \end{pmatrix} = A \checkmark$$

Q. 7. Is $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ orthogonal? What is $|A|$?

$A^T = A$. $A^T A$: Row 1 $\times$ Col 1 = $0 + 1 = 1$; Row 1 $\times$ Col 2 = $0 + 0 = 0$; Row 2 $\times$ Col 1 = 0 ; Row 2 $\times$ Col 2 = $1 + 0 = 1$. So $A^T A = I$ and A is orthogonal.

$|A| = 0 - 1 = -1$, which agrees with $|A| = \pm 1$.

Q. 8. Find the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$.

The 2×2 minors are $\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$, $\begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 0$, $\begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 0$. All are zero, but the 1×1 minor $1 \neq 0$. Rank = 1 (row 2 = $2 \times$ row 1).

Q. 9. Solve $x + 2y = 5$, $3x - y = 1$ by the matrix method.

$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$. $|A| = -1 - 6 = -7 \neq 0$.

$\text{adj } A = \begin{pmatrix} -1 & -2 \\ -3 & 1 \end{pmatrix}$. $(\text{adj } A)B$: Row 1 = $-5 - 2 = -7$; Row 2 = $-15 + 1 = -14$.

$X = (1/-7) \begin{pmatrix} -7 \\ -14 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. So $x = 1$, $y = 2$. Check: $1 + 4 = 5 \checkmark$, $3 - 2 = 1 \checkmark$.

Q. 10. If $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, show that A is idempotent and find A^5 .

A^2 : Row 1 $\times$ Col 1 = $1 \cdot 1 + 1 \cdot 0 = 1$; Row 1 $\times$ Col 2 = $1 \cdot 1 + 1 \cdot 0 = 1$; Row 2 $\times$ Col 1 = 0 ; Row 2 $\times$ Col 2 = 0 . $A^2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = A$, so A is idempotent.

Then every power equals A : $A^5 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$.

UNIT 7

Determinants

ഓരോ square matrix-നും ഒരു single number ആയ determinant ഉണ്ട്; അത് എങ്ങനെ കണ്ടെത്താം (minors, cofactors, expansion), properties ഉപയോഗിച്ച് കണക്ക് എങ്ങനെ ചുരുക്കാം, adjoint, reciprocal, symmetric, skew-symmetric determinants, അവസാനം Cramer's rule ഉപയോഗിച്ച് simultaneous equations solve ചെയ്യുന്നത് എന്നിവയാണ് ഈ unit.

What is a determinant?

Determinants of order 1 and 2

Minors and cofactors

Evaluating a 3×3 determinant

Properties I to IV

Properties V to VIII

Evaluating determinants using properties

Product of two determinants

Adjoint and reciprocal determinants (and the inverse of a matrix)

Symmetric determinants

Skew and skew-symmetric determinants

Cramer's rule for two equations

Cramer's rule for three equations

What is a determinant?

Definition: A determinant is a single number (scalar) associated with a square matrix A , written $\det A$ or $|A|$.

ഒരു square matrix A -യുമായി ബന്ധപ്പെട്ട ഒരു single number (scalar) ആണ് determinant; അതിനെ $\det A$ അല്ലെങ്കിൽ $|A|$ എന്ന് എഴുതുന്നു.

A **determinant** is a single number (scalar) attached to a **square** matrix. It is written $\det A$ or $|A|$. Historically it came from asking when linear equations have a common solution: $a_1x + b_1 = 0$ and $a_2x + b_2 = 0$ share a solution exactly when $a_1b_2 - a_2b_1 = 0$. That expression is the second-order determinant. Similarly, three equations in two unknowns $a_ix + b_iy + c_i = 0$ ($i = 1, 2, 3$) have a common solution when a third-order determinant equals 0. Determinants were used by Seki Kowa and Leibniz around 1683, long before matrices were invented.

മലയാളത്തിൽ

Determinant എന്നത് ഒരു square matrix-ൽ നിന്ന് കിട്ടുന്ന ഒരു single number ആണ്. എല്ലാ subjects-ന്റെയും marks ചേർത്ത് ഒരു total കണക്കാക്കുന്നതുപോലെ, matrix-ലെ എല്ലാ elements-ഉം ഒരു നിശ്ചിത rule അനുസരിച്ച് ചേർന്ന് ഈ ഒരു number ഉണ്ടാകുന്നു. Rows-ഉം columns-ഉം തുല്യമല്ലെങ്കിൽ (square അല്ലെങ്കിൽ) determinant ഇല്ല. Equations-ന് common solution ഉണ്ടോ, matrix-ന് inverse ഉണ്ടോ എന്നൊക്കെ ഈ number പറഞ്ഞുതരും.

$$a_1x + b_1 = 0 \text{ and } a_2x + b_2 = 0 \text{ have a common solution} \Leftrightarrow \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1 = 0$$

A determinant of order n has n rows, n columns and $n!$ terms in its expansion (half with + sign, half with - sign).

Order 2 $\rightarrow 2! = 2$ terms; order 3 $\rightarrow 3! = 6$ terms.

The numbers inside are the **elements**; horizontal lines are **rows**, vertical lines are **columns**.

EXAMPLE

Do $2x - 6 = 0$ and $5x - 15 = 0$ have a common solution? Check with a determinant.

Here $a_1 = 2$, $b_1 = -6$, $a_2 = 5$, $b_2 = -15$.

$$\begin{vmatrix} 2 & -6 \\ 5 & -15 \end{vmatrix} = 2(-15) - 5(-6) = -30 + 30 = 0.$$

The determinant is 0, so yes: both equations give $x = 3$.

Exam tip: A determinant exists only for a square matrix. If A is 2×3 , " $|A|$ " has no meaning. For a square matrix the determinant is unique.

Determinants of order 1 and 2

Order 1: The determinant of $[a_{11}]$ is a_{11} itself.

$[a_{11}]$ -ന്റെ determinant a_{11} തന്നെയാണ്.

Order 2: For a 2×2 matrix $[a_{ij}]$, $|A| = a_{11}a_{22} - a_{21}a_{12}$ (main-diagonal product minus other-diagonal product).

2×2 matrix $[a_{ij}]$ -ന് $|A| = a_{11}a_{22} - a_{21}a_{12}$ (main diagonal-ന്റെ product-ൽ നിന്ന് മറ്റേ diagonal-ന്റെ product കുറച്ചത്).

For a 1×1 matrix $A = [a_{11}]$, $|A| = a_{11}$. Careful: here $|\cdot|$ means determinant, not modulus, so $|-8| = -8$ as a determinant. For a 2×2 matrix, multiply down the main diagonal and subtract the product of the other diagonal.

മലയാളത്തിൽ

1×1 matrix-ന്റെ determinant ആ number തന്നെ; $|-3| = -3$ ആണ്, 3 അല്ല, കാരണം ഇവിടെ $||$ modulus അല്ല, determinant ആണ്. 2×2 -ൽ "ഇടത്തുനിന്ന് താഴോട്ടുള്ള diagonal ഗുണിക്കുക, മറ്റേ diagonal ഗുണിച്ച് കുറയ്ക്കുക" എന്ന ഒരു rule മാത്രം ഓർത്താൽ മതി.

$A = [a_{11}] \Rightarrow |A| = a_{11}$ e.g. $|[5]| = 5$, $|[-3]| = -3$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

EXAMPLE (IGNOU EXAMPLE 1)

Evaluate (i) $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ (ii) $\begin{vmatrix} 3 & 5 \\ 8 & 9 \end{vmatrix}$ (iii) $\begin{vmatrix} x^2 & x^2 + 1 \\ x & x + 1 \end{vmatrix}$

(i) $ad - bc$.

(ii) $3 \times 9 - 8 \times 5 = 27 - 40 = -13$.

(iii) $x^2(x + 1) - x(x^2 + 1) = x^3 + x^2 - x^3 - x = x^2 - x$.

EXAMPLE (CHECK YOUR PROGRESS 1, Q1)

Find x if (i) $\begin{vmatrix} x & 7 \\ 9 & x+2 \end{vmatrix} = 0$ (ii) $\begin{vmatrix} x & x^2 \\ 15 & 5 \end{vmatrix} = 0$.

(i) $x(x+2) - 63 = 0 \Rightarrow x^2 + 2x - 63 = 0 \Rightarrow (x+9)(x-7) = 0 \Rightarrow x = 7$ or $x = -9$.

(ii) $5x - 15x^2 = 0 \Rightarrow 5x(1 - 3x) = 0 \Rightarrow x = 0$ or $x = 1/3$.

Minors and cofactors

Minor: The minor M_{ij} of a_{ij} is the determinant left after deleting the i -th row and j -th column.

i -ാം row-ഉം j -ാം column-ഉം ഒഴിവാക്കിയ ശേഷം ബാക്കിയുള്ള determinant ആണ് a_{ij} -യുടെ minor M_{ij} .

Cofactor: $C_{ij} = (-1)^{i+j} M_{ij}$, the minor with its place sign attached.

$C_{ij} = (-1)^{i+j} M_{ij}$; അതായത് position-ന്റെ sign ചേർത്ത minor.

The **minor** M_{ij} of the element a_{ij} is the determinant left after deleting the i -th row and j -th column. So in an n -th order determinant every minor has order $n - 1$. The **cofactor** C_{ij} is the minor with a sign attached: $C_{ij} = (-1)^{i+j} M_{ij}$. The sign follows a chessboard pattern that starts with + at position (1, 1).

മലയാളത്തിൽ

ഒരു element-ന്റെ minor കിട്ടാൻ, ആ element ഇരിക്കുന്ന row-ഉം column-ഉം "വെട്ടിക്കളയുക"; ബാക്കിയുള്ള ചെറിയ determinant ആണ് minor. Cofactor എന്നാൽ അതേ minor-ന് + അല്ലെങ്കിൽ - sign കൊടുത്തത്. Sign ചെസ്സ് ബോർഡിലെ കറുപ്പും വെളുപ്പും കളങ്ങൾ പോലെ മാറിമാറി വരും: (1, 1)-ൽ +, അടുത്ത കളത്തിൽ -, അങ്ങനെ. $i + j$ even ആണെങ്കിൽ +, odd ആണെങ്കിൽ -.

M_{ij} = determinant after deleting row i and column j

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Sign pattern for 3×3 :

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

IGNOU also writes A_i, B_i, C_i for the cofactors of a_i, b_i, c_i in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

EXAMPLE (IGNOU EXAMPLE 2(I) AND CYP 1, Q2(I))

Find all minors and cofactors of $A = \begin{pmatrix} 2 & 5 \\ 4 & -7 \end{pmatrix}$.

Deleting row 1, column 1 leaves -7 , so $M_{11} = -7$. Likewise $M_{12} = 4$, $M_{21} = 5$, $M_{22} = 2$.

$C_{11} = (+1)(-7) = -7$, $C_{12} = (-1)(4) = -4$, $C_{21} = (-1)(5) = -5$, $C_{22} = (+1)(2) = 2$.

EXAMPLE (IGNOU EXAMPLE 2(II) AND CYP 1, Q2(II))

Find all minors and cofactors of $A = \begin{pmatrix} -3 & 4 & -2 \\ 6 & 5 & 7 \\ -8 & 9 & 1 \end{pmatrix}$.

$$M_{11} = \begin{vmatrix} 5 & 7 \\ 9 & 1 \end{vmatrix} = 5 - 63 = -58, M_{12} = \begin{vmatrix} 6 & 7 \\ -8 & 1 \end{vmatrix} = 6 + 56 = 62, M_{13} = \begin{vmatrix} 6 & 5 \\ -8 & 9 \end{vmatrix} = 54 + 40 = 94$$

$$M_{21} = \begin{vmatrix} 4 & -2 \\ 9 & 1 \end{vmatrix} = 4 + 18 = 22, M_{22} = \begin{vmatrix} -3 & -2 \\ -8 & 1 \end{vmatrix} = -3 - 16 = -19, M_{23} = \begin{vmatrix} -3 & 4 \\ -8 & 9 \end{vmatrix} = -27 + 32 = 5$$

$$M_{31} = \begin{vmatrix} 4 & -2 \\ 5 & 7 \end{vmatrix} = 28 + 10 = 38, M_{32} = \begin{vmatrix} -3 & -2 \\ 6 & 7 \end{vmatrix} = -21 + 12 = -9, M_{33} = \begin{vmatrix} -3 & 4 \\ 6 & 5 \end{vmatrix} = -15 - 24 = -39$$

Apply the sign pattern: $C_{11} = -58$, $C_{12} = -62$, $C_{13} = 94$, $C_{21} = -22$, $C_{22} = -19$, $C_{23} = -5$, $C_{31} = 38$, $C_{32} = 9$, $C_{33} = -39$.

Bonus check: $|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = (-3)(-58) + 4(-62) + (-2)(94) = 174 - 248 - 188 = -262$.

EXAMPLE (CYP 1, Q3)

For $A = \begin{pmatrix} 5 & 6 & -2 \\ -3 & 8 & 9 \\ 7 & 10 & -4 \end{pmatrix}$, find the minor and cofactor of a_{12} , a_{23} , a_{31} , a_{13} .

$$M_{12} = \begin{vmatrix} -3 & 9 \\ 7 & -4 \end{vmatrix} = 12 - 63 = -51, C_{12} = -(-51) = 51.$$

$$M_{23} = \begin{vmatrix} 5 & 6 \\ 7 & 10 \end{vmatrix} = 50 - 42 = 8, C_{23} = -8.$$

$$M_{31} = \begin{vmatrix} 6 & -2 \\ 8 & 9 \end{vmatrix} = 54 + 16 = 70, C_{31} = 70.$$

$$M_{13} = \begin{vmatrix} -3 & 8 \\ 7 & 10 \end{vmatrix} = -30 - 56 = -86, C_{13} = -86.$$

Exam tip: The most common mistake is forgetting the sign: minor and cofactor are equal only when $i + j$ is even. In the answer key for CYP 1, Q2 the unit says "Using Example 11"; the minors actually come from Example 2.

Evaluating a 3×3 determinant

Definition: A determinant equals the sum of the products of the elements of any one row (or column) with their cofactors, e.g. $|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$.

ഏതെങ്കിലും ഒരു row-ലെ (അല്ലെങ്കിൽ column-ലെ) elements-നെ അവയുടെ cofactors കൊണ്ട് ഗുണിച്ച് കൂട്ടിയതാണ് determinant-ന്റെ value; ഉദാ. $|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$.

Pick any row (or column). Multiply each element by its cofactor and add. Every row and every column gives the same value. Choose the line with the most zeros to save work. For 3×3 only, there is also a shortcut (the diagonal or "Sarrus" rule): write the first two columns again to the right, add the three down-right diagonal products and subtract the three up-right diagonal products.

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3×3 determinant കാണാൻ ഏതെങ്കിലും ഒരു row അല്ലെങ്കിൽ column തിരഞ്ഞെടുക്കുക; ഓരോ element-നെയും അതിന്റെ cofactor കൊണ്ട് ഗുണിച്ച് കൂട്ടുക. ഏത് row എടുത്താലും ഉത്തരം ഒന്നുതന്നെ. കൂടുതൽ zero ഉള്ള row/column എടുത്താൽ പണി പകുതിയാകും, കാരണം zero-നെ ഗുണിക്കുന്ന term-കൾ കണക്കാക്കേണ്ടതേയില്ല. Sarrus rule 3×3 -ന് മാത്രമാണ്, 4×4 -ന് പറ്റില്ല.

Expansion along R_1 :

$$|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

$$= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

Diagonal rule:
$$\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = aqz + brx + cpy - ary - bpz - cqx$$

EXAMPLE (IGNOU EXAMPLE 3)

Find the value of $\Delta = \begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{vmatrix}$ by expanding along R_1 and along C_1 .

$$\text{Along } R_1: \Delta = 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} + 7 \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix} = 3(1 - 6) - 4(2 - 21) + 7(4 - 7) = -15 + 76 - 21 = 40.$$

$$\text{Along } C_1: \Delta = 3 \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 7 \\ 2 & 1 \end{vmatrix} + 7 \begin{vmatrix} 4 & 7 \\ 1 & 3 \end{vmatrix} = 3(1 - 6) - 2(4 - 14) + 7(12 - 7) = -15 + 20 + 35 = 40.$$

Same answer, as expected.

EXAMPLE (IGNOU EXAMPLE 4)

Evaluate (i) $\begin{vmatrix} 3 & 2 & -1 \\ 5 & 4 & 6 \\ -3 & 1 & 7 \end{vmatrix}$ (ii) $\begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix}$ (iii) $\begin{vmatrix} 3 & 1 & 2 \\ 0 & 9 & 6 \\ 0 & 5 & 4 \end{vmatrix}$

(i) Along R_1 : $3(28 - 6) - 2(35 + 18) + (-1)(5 + 12) = 66 - 106 - 17 = -57$.

Check with the diagonal rule: $(3 \cdot 4 \cdot 7 + 2 \cdot 6 \cdot (-3) + (-1) \cdot 5 \cdot 1) - ((-1) \cdot 4 \cdot (-3) + 3 \cdot 6 \cdot 1 + 2 \cdot 5 \cdot 7) = (84 - 36 - 5) - (12 + 18 + 70) = 43 - 100 = -57$. ✓

(ii) Along R_1 : $2(2 - 4) - 1(1 - 4) + 2(2 - 4) = -4 + 3 - 4 = -5$.

(iii) C_1 has two zeros, so expand along C_1 : $3(36 - 30) - 0 + 0 = 18$.

EXAMPLE (CYP 1, Q4)

Show that $|A| = 0$ for $A = \begin{pmatrix} 1 & 2 & 4 \\ -3 & 5 & 1 \\ 2 & 4 & 8 \end{pmatrix}$.

Along R_1 : $1(40 - 4) - 2(-24 - 2) + 4(-12 - 10) = 36 + 52 - 88 = 0$.

Quicker: $R_3 = 2R_1$, so the rows are proportional and $|A| = 0$ (Property III, next topic).

Exam tip: In the unit's column expansion of Example 3 the last line is printed as "-15 + 20 - 35 = 40"; the correct line is $-15 + 20 + 35 = 40$, since $7(12 - 7) = +35$.

Properties I to IV

Definition: Properties of determinants are rules, valid for every order, that give how the value changes when lines are transposed, interchanged, identical or proportional, or multiplied by a constant.

Lines transpose ചെയ്യുമ്പോഴോ, പരസ്പരം മാറ്റുമ്പോഴോ, identical അല്ലെങ്കിൽ proportional ആകുമ്പോഴോ, ഒരു constant കൊണ്ട് ഗുണിക്കുമ്പോഴോ determinant-ന്റെ value എങ്ങനെ മാറുന്നു എന്ന് പറയുന്ന, എല്ലാ order-നും ബാധകമായ rules ആണ് properties of determinants.

Direct expansion is slow and error-prone for big determinants. The properties let you simplify first. They are proved for order 3 but hold for every order.

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Properties എന്നത് കണക്ക് ചുരുക്കാനുള്ള "shortcut-കൾ" ആണ്. ഒരു വലിയ ചാക്ക് അരി അളക്കും മുമ്പ് കല്ലും പതിരും കളയുന്നതുപോലെ, expand ചെയ്യും മുമ്പ് zeros ഉണ്ടാക്കി determinant ലളിതമാക്കുന്നു. Rows-ഉം columns-ഉം മാറ്റിയാൽ value മാറില്ല; രണ്ട് rows swap ചെയ്താൽ sign മാത്രം മാറും; രണ്ട് rows ഒരുപോലെ ആണെങ്കിൽ value zero; ഒരു row-ൽ common factor ഉണ്ടെങ്കിൽ അത് പുറത്തെടുക്കാം.

I. Changing rows into columns (transpose) does not change the value.

II. Interchanging two rows (or two columns) multiplies the value by -1 .

III. If two rows (or columns) are identical or proportional, the value is 0.

IV. Multiplying every element of one row (or column) by k multiplies the value by k . So a common factor of a line can be taken outside.

Consequence of IV:
$$\begin{vmatrix} k_1 a_1 & k_1 b_1 & k_1 c_1 \\ k_2 a_2 & k_2 b_2 & k_2 c_2 \\ k_3 a_3 & k_3 b_3 & k_3 c_3 \end{vmatrix} = k_1 k_2 k_3 \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

For an $n \times n$ matrix: $|kA| = k^n |A|$ (k multiplies every one of the n rows).

EXAMPLE (PROPERTIES I AND II, IGNOU EXAMPLE 5)

$$\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 3 \end{vmatrix} \quad (\text{Property I, transpose}).$$

$$\text{Swap } C_1 \text{ and } C_2 := - \begin{vmatrix} 3 & 1 & 0 \\ 4 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix}. \text{ Now swap } R_1 \text{ and } R_2 := + \begin{vmatrix} 4 & 2 & 0 \\ 3 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix}.$$

Check: the original, expanded along C_3 , is $3(4 - 6) = -6$; the last one is $3(4 - 6) = -6$. ✓

EXAMPLE (PROPERTY III, IGNOU EXAMPLES 6 AND 7)

$$\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 7 \\ 1 & 3 & 0 \end{vmatrix} = 0 \text{ because } R_1 \text{ and } R_3 \text{ are identical.}$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 0 \text{ because } C_2 = 2 \times C_1 \text{ (proportional columns).}$$

EXAMPLE (PROPERTY IV, IGNOU EXAMPLE 8)

$$\begin{vmatrix} 2 & 10 & 16 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix} = 2 \begin{vmatrix} 1 & 5 & 8 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix}, \text{ taking 2 out of } R_1.$$

Check: the right-hand determinant is $1(77 - 4) - 5(33 - 5) + 8(12 - 35) = 73 - 140 - 184 = -251$, and $2 \times (-251) = -502$, which is the value of the left side: $2(77 - 4) - 10(33 - 5) + 16(12 - 35) = 146 - 280 - 368 = -502$. ✓

Exam tip: Taking k out of a determinant uses only one row or column. Taking k out of a matrix (kA) takes it out of every row, so $|3A| = 27|A|$ for a 3×3 matrix, not $3|A|$.

Properties V to VIII

Definition: The elementary operation $R_i \rightarrow R_i + kR_j$ (or $C_i \rightarrow C_i + kC_j$), $i \neq j$, adds k times one line to another parallel line; it leaves the determinant unchanged.

$R_i \rightarrow R_i + kR_j$ (അല്ലെങ്കിൽ $C_i \rightarrow C_i + kC_j$), $i \neq j$, എന്ന elementary operation ഒരു line-ന്റെ k മടങ്ങ് മറ്റൊരു parallel line-നോട് കൂട്ടുന്നു; ഇത് determinant-ന്റെ value മാറ്റില്ല.

These properties are the real "zero-making" tools. Property VI is the one you will use in almost every problem: you may add a multiple of one row to another row without changing the value. Rows are written R_i , columns C_j ; " $R_2 \rightarrow R_2 - R_1$ " means "replace row 2 by row 2 minus row 1".

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Property VI ആണ് ഏറ്റവും ഉപയോഗമുള്ളത്: ഒരു row-ൽ നിന്ന് മറ്റൊരു row-ന്റെ k മടങ്ങ് കൂട്ടുകയോ കുറയ്ക്കുകയോ ചെയ്താൽ value മാറില്ല. ഇതുപയോഗിച്ച് നമുക്ക് ഇഷ്യമുള്ളിടത് zeros ഉണ്ടാക്കാം. ഒരു row മുഴുവൻ രണ്ട് numbers-ന്റെ sum ആണെങ്കിൽ determinant-നെ രണ്ട് determinants ആയി പിരിക്കാം (Property V). ഒരു row മുഴുവൻ zero ആണെങ്കിൽ value zero (Property VIII).

V. If every element of one line is a sum of two terms, the determinant splits into a sum of two determinants:

$$\begin{vmatrix} a_1 + \alpha_1 & b_1 + \beta_1 & c_1 + \delta_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} \alpha_1 & \beta_1 & \delta_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

VI. $R_i \rightarrow R_i + kR_j$ (or $C_i \rightarrow C_i + kC_j$), $i \neq j$, leaves the value unchanged.

VII. $|A'| = |A|$ (a matrix and its transpose have equal determinants).

VIII. If every element of a row (or column) is 0, the value is 0.

EXAMPLE (PROPERTY V, IGNOU EXAMPLE 9)

$$\text{Verify: } \Delta = \begin{vmatrix} 8+2 & 2+5 & 5+4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} = \Delta_1 + \Delta_2 \text{ where } \Delta_1 = \begin{vmatrix} 8 & 2 & 5 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix}, \Delta_2 = \begin{vmatrix} 2 & 5 & 4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix}.$$

All three share rows 2 and 3, so the three 2×2 cofactors of row 1 are the same: $(63 - 11) = 52$, $(28 - 5) = 23$, $(44 - 45) = -1$.

$$\Delta_1 = 8(52) - 2(23) + 5(-1) = 416 - 46 - 5 = 365.$$

$$\Delta_2 = 2(52) - 5(23) + 4(-1) = 104 - 115 - 4 = -15.$$

$$\Delta = 10(52) - 7(23) + 9(-1) = 520 - 161 - 9 = 350 = 365 + (-15). \checkmark$$

EXAMPLE (WHY PROPERTY VI WORKS, IGNOU EXAMPLE 10)

$$\begin{vmatrix} a_1 + kb_1 & b_1 & c_1 \\ a_2 + kb_2 & b_2 & c_2 \\ a_3 + kb_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + k \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix} \text{ (Properties V and IV).}$$

The second determinant has $C_1 = C_2$, so it is 0 (Property III). Hence the value is unchanged.

Exam tip: In $R_i \rightarrow R_i + kR_j$, the row you change (R_i) must not be multiplied by anything. " $R_1 \rightarrow 2R_1 + R_2$ " doubles the value. Also, the unit's worked solutions cite "P7" for a zero column; in the unit's own numbering that is Property VIII.

Evaluating determinants using properties

Standard plan: (1) look for a row/column operation that makes a whole line equal (often $C_1 \rightarrow C_1 + C_2 + C_3$); (2) take the common factor out; (3) subtract rows to create zeros; (4) expand along the line with the zeros. The nine parts of IGNOU Example 11 cover every pattern asked in exams.

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Properties ഉപയോഗിക്കുന്ന രീതി ഒരു ചെറിയ recipe പോലെ ആണ്: ആദ്യം എല്ലാ columns-ഉം കൂട്ടി നോക്കുക, ഒരേ expression വരുന്നുണ്ടോ എന്ന്; വന്നാൽ അത് common ആയി പുറത്തെടുക്കുക; പിന്നെ rows തമ്മിൽ കുറച്ച് zeros ഉണ്ടാക്കുക; അവസാനം zero കൂടുതലുള്ള line-ൽ expand ചെയ്യുക. ഈ steps മനഃപാഠമാക്കിയാൽ proof ചോദ്യങ്ങൾ എളുപ്പമാകും.

EXAMPLE (IGNOU EXAMPLE 11, PARTS (I) TO (V): VALUE 0)

$$(i) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} : C_1 \rightarrow C_1 + C_2 + C_3 \text{ makes every entry of } C_1 \text{ equal to 0, so } \Delta = 0 \text{ (Property VIII).}$$

$$(ii) \begin{vmatrix} -3 & 5 & -2 \\ 8 & 9 & -17 \\ 3 & -6 & 3 \end{vmatrix} : \text{each row adds to 0 } (-3 + 5 - 2, 8 + 9 - 17, 3 - 6 + 3), \text{ so } C_1 \rightarrow C_1 + C_2 + C_3 \text{ gives a zero column. } \Delta = 0.$$

$$(iii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix} : R_3 \rightarrow R_3 + R_2 \text{ makes } R_3 = (a+b+c, a+b+c, a+b+c). \text{ Take } (a+b+c) \text{ out: } R_3 \text{ becomes } (1, 1, 1) = R_1, \text{ so } \Delta = 0.$$

$$(iv) \begin{vmatrix} 3 & x & xyz \\ 3 & y & xyz \\ 3 & z & xyz \end{vmatrix} : \text{take 3 from } C_1 \text{ and } xyz \text{ from } C_3 : \Delta = 3xyz \begin{vmatrix} 1 & x & 1 \\ 1 & y & 1 \\ 1 & z & 1 \end{vmatrix} = 0 \text{ (} C_1 = C_3 \text{)}.$$

$$(v) \begin{vmatrix} 2 & 3 & 30 \\ 5 & 4 & 54 \\ 6 & 1 & 42 \end{vmatrix} : \text{take 6 from } C_3, \text{ giving } C_3 = (5, 9, 7). \text{ Then } C_3 \rightarrow C_3 - C_1 - C_2 \text{ gives } (0, 0, 0). \Delta = 0.$$

EXAMPLE (IGNOU EXAMPLE 11(VI): VANDERMONDE DETERMINANT)

Show $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x - y)(y - z)(z - x).$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1: \begin{vmatrix} 1 & x & x^2 \\ 0 & y - x & y^2 - x^2 \\ 0 & z - x & z^2 - x^2 \end{vmatrix}.$$

Take $(y - x)$ from R_2 and $(z - x)$ from R_3 : $(y - x)(z - x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y + x \\ 0 & 1 & z + x \end{vmatrix}.$

Expand along C_1 : $(y - x)(z - x)[(z + x) - (y + x)] = (y - x)(z - x)(z - y).$

Since $(y - x)(z - y) = (x - y)(y - z)$, this equals $(x - y)(y - z)(z - x).$

Check with $x = 1, y = 2, z = 3$: the determinant is 2 and $(-1)(-1)(2) = 2. \checkmark$

EXAMPLE (IGNOU EXAMPLE 11(VII))

Show $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3.$

$$C_1 \rightarrow C_1 + C_2 + C_3, \text{ take } (a + b + c) \text{ out: } (a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}.$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1: (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & c - b & a - c \\ 0 & a - b & b - c \end{vmatrix}.$$

Expand along C_1 : $(a + b + c)[(c - b)(b - c) - (a - c)(a - b)] = (a + b + c)(ab + bc + ca - a^2 - b^2 - c^2) = 3abc - a^3 - b^3 - c^3.$

EXAMPLE (IGNOU EXAMPLE 11(VIII))

Show
$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z).$$

$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$, then take $(y-x), (z-x)$ out (using $y^3 - x^3 = (y-x)(y^2 + xy + x^2)$): $(y-x)(z-x)$

$$\begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^3 & x^2 + xy + y^2 & x^2 + xz + z^2 \end{vmatrix}.$$

$C_3 \rightarrow C_3 - C_2$: the bottom entry becomes $xz - xy + z^2 - y^2 = (z-y)(x+y+z)$, the middle becomes 0.

Expand along R_1 : $(y-x)(z-x)(z-y)(x+y+z) = (x-y)(y-z)(z-x)(x+y+z)$.

Check with $x = 1, y = 2, z = 3$: determinant = 12 and $(-1)(-1)(2)(6) = 12$. $\checkmark$

EXAMPLE (IGNOU EXAMPLE 11(IX))

Evaluate
$$\begin{vmatrix} 2x+3 & x & x \\ x & 2x+3 & x \\ x & x & 2x+3 \end{vmatrix}.$$

$C_1 \rightarrow C_1 + C_2 + C_3$: every entry of C_1 is $4x+3$. Take it out: $(4x+3) \begin{vmatrix} 1 & x & x \\ 1 & 2x+3 & x \\ 1 & x & 2x+3 \end{vmatrix}.$

$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$: $(4x+3) \begin{vmatrix} 1 & x & x \\ 0 & x+3 & 0 \\ 0 & 0 & x+3 \end{vmatrix} = (4x+3)(x+3)^2.$

Check with $x = 2$: direct value 275 and $11 \times 25 = 275$. $\checkmark$

EXAMPLE (CYP 2, Q1)

Without expanding, prove
$$\begin{vmatrix} ab & 1 & c(a+b) \\ bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \end{vmatrix} = 0.$$

$C_3 \rightarrow C_3 + C_1$: each entry becomes $ab + bc + ca$. Take it out: $(ab + bc + ca) \begin{vmatrix} ab & 1 & 1 \\ bc & 1 & 1 \\ ca & 1 & 1 \end{vmatrix} = 0$, since $C_2 = C_3$.

Exam tip: When a proof question says "using properties", full expansion at the first step loses marks. Write each operation (e.g. " $C_1 \rightarrow C_1 + C_2 + C_3$ ") next to the step; examiners look for it. Quickly test your final factorised answer with small numbers like $x = 1, y = 2, z = 3$.

Product of two determinants

Definition: Two determinants of the same order n multiply (row by column) to give a determinant of order n , and $\det(AB) = \det(A) \cdot \det(B)$.

ഒരേ order n ഉള്ള രണ്ട് determinants-നെ (row by column) ഗുണിച്ചാൽ order n ഉള്ള ഒരു determinant കിട്ടും; $\det(AB) = \det(A) \cdot \det(B)$.

Two determinants can be multiplied only if they have the same order; the product is a determinant of that order. You multiply them like matrices (row of the first by column of the second). Because $|A| = |A'|$, you may also use row-by-row, column-by-column or column-by-row multiplication and get the same value. The key fact is $\det(AB) = \det(A) \cdot \det(B)$.

മലയാളത്തിൽ

രണ്ട് determinants ഗുണിക്കാൻ അവയുടെ order ഒന്നായിരിക്കണം. Matrix multiplication പോലെ തന്നെ, ആദ്യത്തേതിന്റെ row-നെ രണ്ടാമത്തേതിന്റെ column കൊണ്ട് ഗുണിച്ച് കൂട്ടുന്നു. പക്ഷേ ഉത്തരത്തിന്റെ value മാത്രമാണ് വേണ്ടതെങ്കിൽ, രണ്ടിന്റെയും values കണ്ടെത്തി നേരിട്ട് ഗുണിച്ചാൽ മതി: $|AB| = |A| \cdot |B|$.

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} c_1 & d_1 \\ c_2 & d_2 \end{vmatrix} = \begin{vmatrix} a_1c_1 + b_1c_2 & a_1d_1 + b_1d_2 \\ a_2c_1 + b_2c_2 & a_2d_1 + b_2d_2 \end{vmatrix} \text{ (row by column)}$$

$$|AB| = |A| \cdot |B|$$

Four allowed rules: row $\times$ column, row $\times$ row, column $\times$ column, column $\times$ row.

EXAMPLE (IGNOU'S WORKED PRODUCT, CORRECTED)

Find $\begin{vmatrix} 4 & 5 & 9 \\ 1 & 2 & 3 \\ 3 & 0 & 2 \end{vmatrix} \times \begin{vmatrix} 2 & 1 & 5 \\ 3 & 2 & 1 \\ 1 & 6 & 4 \end{vmatrix}$ by the row-by-column rule.

Row 1 $\times$ columns: $4 \cdot 2 + 5 \cdot 3 + 9 \cdot 1 = 32$; $4 \cdot 1 + 5 \cdot 2 + 9 \cdot 6 = 68$; $4 \cdot 5 + 5 \cdot 1 + 9 \cdot 4 = 61$.

Row 2 $\times$ columns: $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 1 = 11$; $1 \cdot 1 + 2 \cdot 2 + 3 \cdot 6 = 23$; $1 \cdot 5 + 2 \cdot 1 + 3 \cdot 4 = 19$.

Row 3 $\times$ columns: $3 \cdot 2 + 0 \cdot 3 + 2 \cdot 1 = 8$; $3 \cdot 1 + 0 \cdot 2 + 2 \cdot 6 = 15$; $3 \cdot 5 + 0 \cdot 1 + 2 \cdot 4 = 23$.

$$\text{Product} = \begin{vmatrix} 32 & 68 & 61 \\ 11 & 23 & 19 \\ 8 & 15 & 23 \end{vmatrix}.$$

Check the value: first determinant $= 4(4 - 0) - 5(2 - 9) + 9(0 - 6) = 16 + 35 - 54 = -3$; second $= 2(8 - 6) - 1(12 - 1) + 5(18 - 2) = 4 - 11 + 80 = 73$. So the product determinant equals $(-3)(73) = -219$.

EXAMPLE

Using row-by-row multiplication, find $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \times \begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix}$.

Row-by-row: entry $(i, j) = (\text{row } i \text{ of first}) \cdot (\text{row } j \text{ of second})$.

$(1, 1) = 1 \cdot 2 + 2 \cdot 0 = 2$; $(1, 2) = 1 \cdot 1 + 2 \cdot 3 = 7$; $(2, 1) = 3 \cdot 2 + 4 \cdot 0 = 6$; $(2, 2) = 3 \cdot 1 + 4 \cdot 3 = 15$.

$$\text{Product} = \begin{vmatrix} 2 & 7 \\ 6 & 15 \end{vmatrix} = 30 - 42 = -12, \text{ and directly } (4 - 6)(6 - 0) = (-2)(6) = -12. \checkmark$$

Exam tip: In the unit's worked product the third row of the first determinant is printed as $(3, 0, 1)$, but the working uses $3 \cdot 2 + 0 \cdot 3 + 2 \cdot 1$, i.e. the row $(3, 0, 2)$. With $(3, 0, 2)$ the result 32, 68, 61 / 11, 23, 19 / 8, 15, 23 is correct; that is the version shown here.

Adjoint and reciprocal determinants (and the inverse of a matrix)

Adjoint determinant: The determinant Δ' formed by replacing every element of Δ by its cofactor.

Δ -യിലെ ഓരോ element-നെയും അതിന്റെ cofactor കൊണ്ട് മാറ്റി കിട്ടുന്ന determinant Δ' .

Reciprocal determinant: For $\Delta \neq 0$, the determinant formed by dividing every element of the adjoint of Δ by Δ .

$\Delta \neq 0$ ആണെങ്കിൽ, Δ -യുടെ adjoint-ലെ ഓരോ element-നെയും Δ കൊണ്ട് ഹരിച്ച് കിട്ടുന്ന determinant.

Adjoint matrix and inverse: $\text{adj } A$ is the transpose of the cofactor matrix of A , and $A^{-1} = (1/|A|) \text{adj } A$ when $|A| \neq 0$.

A -യുടെ cofactor matrix-ന്റെ transpose ആണ് $\text{adj } A$; $|A| \neq 0$ ആണെങ്കിൽ $A^{-1} = (1/|A|) \text{adj } A$.

The **adjoint** (adjugate) of a determinant Δ is the determinant Δ' formed by replacing every element by its cofactor. The **reciprocal** of Δ ($\Delta \neq 0$) is obtained by dividing every element of the adjoint by Δ . For an $n \times n$ matrix A , the adjoint matrix $\text{adj } A$ is the **transpose** of the cofactor matrix, and $A^{-1} = (1/|A|) \text{adj } A$ when $|A| \neq 0$. For determinant values the transpose makes no difference (Property I), but for the matrix inverse it matters.

മലയാളത്തിൽ

Adjoint ഉണ്ടാക്കാൻ ഓരോ element-നും പകരം അതിന്റെ cofactor എഴുതുക. Matrix-ന്റെ $\text{adj } A$ -ക്ക് ആ cofactor matrix-നെ transpose ചെയ്യണം (rows columns ആക്കണം). പിന്നെ $|A|$ കൊണ്ട് ഹരിച്ചാൽ inverse കിട്ടും. $|A| = 0$ ആണെങ്കിൽ ഹരിക്കാൻ പറ്റില്ല, അതുകൊണ്ട് inverse ഇല്ല; അത്തരം matrix-നെ singular എന്ന് പറയും. $A \times A^{-1} = I$ എന്ന് ഗുണിച്ച് നോക്കി ഉത്തരം ശരിയാണോ എന്ന് എപ്പോഴും check ചെയ്യാം.

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow \text{adjoint } \Delta' = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$$

Value of the adjoint of a third-order determinant: $\Delta' = \Delta^2$ (in general Δ^{n-1})

Reciprocal of $\Delta = (1/\Delta^3) \cdot \Delta' = \Delta^2/\Delta^3 = 1/\Delta = \Delta^{-1}$

Matrix: $\text{adj } A = (\text{cofactor matrix})^T$, $A \cdot \text{adj } A = \text{adj } A \cdot A = |A| I$

$A^{-1} = (1/|A|) \text{adj } A$, exists only if $|A| \neq 0$ (A non-singular)

EXAMPLE (IGNOU EXAMPLES 14 AND 15)

Find the adjoint and reciprocal of $\Delta = \begin{vmatrix} 0 & 1 & 2 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix}$.

Here $a_1 = 0, b_1 = 1, c_1 = 2; a_2 = 2, b_2 = 0, c_2 = 1; a_3 = 1, b_3 = 2, c_3 = 0$.

$$A_1 = + \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = 0 - 2 = -2, B_1 = - \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -(0 - 1) = 1, C_1 = + \begin{vmatrix} 2 & 0 \\ 1 & 2 \end{vmatrix} = 4$$

$$A_2 = - \begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} = -(0 - 4) = 4, B_2 = + \begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix} = -2, C_2 = - \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} = -(0 - 1) = 1$$

$$A_3 = + \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 1, B_3 = - \begin{vmatrix} 0 & 2 \\ 2 & 1 \end{vmatrix} = -(0 - 4) = 4, C_3 = + \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -2$$

$$\text{Adjoint } \Delta' = \begin{vmatrix} -2 & 1 & 4 \\ 4 & -2 & 1 \\ 1 & 4 & -2 \end{vmatrix}.$$

$$\Delta = a_1 A_1 + b_1 B_1 + c_1 C_1 = 0(-2) + 1(1) + 2(4) = 9 \neq 0.$$

$$\text{Reciprocal} = \begin{vmatrix} -2/9 & 1/9 & 4/9 \\ 4/9 & -2/9 & 1/9 \\ 1/9 & 4/9 & -2/9 \end{vmatrix}, \text{ whose value is } 1/9.$$

$$\text{Check } \Delta' = \Delta^2: \text{expanding } \Delta' \text{ along } R_1, -2(4 - 4) - 1(-8 - 1) + 4(16 + 2) = 0 + 9 + 72 = 81 = 9^2. \checkmark$$

EXAMPLE (INVERSE OF A MATRIX)

Find A^{-1} for $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$.

Step 1: $|A|$ along $R_1 = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = -24 + 40 - 15 = 1 \neq 0$, so A^{-1} exists.

Step 2: cofactors. $C_{11} = +(0 - 24) = -24$, $C_{12} = -(0 - 20) = 20$, $C_{13} = +(0 - 5) = -5$; $C_{21} = -(0 - 18) = 18$, $C_{22} = +(0 - 15) = -15$, $C_{23} = -(6 - 10) = 4$; $C_{31} = +(8 - 3) = 5$, $C_{32} = -(4 - 0) = -4$, $C_{33} = +(1 - 0) = 1$.

Step 3: cofactor matrix $\begin{pmatrix} -24 & 20 & -5 \\ 18 & -15 & 4 \\ 5 & -4 & 1 \end{pmatrix}$, so $\text{adj } A = \text{its transpose} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$.

Step 4: $A^{-1} = (1/1) \text{adj } A = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$.

Check one entry of AA^{-1} : row 1 of $A \times$ column 1 $= 1(-24) + 2(20) + 3(-5) = 1$; row 1 $\times$ column 2 $= 18 - 30 + 12 = 0$. $\checkmark$

Exam tip: Forgetting to transpose the cofactor matrix is the classic error in finding A^{-1} . In Example 14 the unit prints $a_3 = 3$; from the determinant it is $a_3 = 1$ (the cofactors in the unit are computed with 1, so they are right). Also, the unit's note " $\Delta^T = \Delta^2$ " means the adjoint's value Δ' equals Δ^2 .

Symmetric determinants

Definition: A determinant $|a_{ij}|$ is symmetric if $a_{ij} = a_{ji}$ for all i, j , i.e. entries are mirror images across the leading diagonal.

എല്ലാ i, j -നും $a_{ij} = a_{ji}$ ആണെങ്കിൽ determinant $|a_{ij}|$ symmetric ആണ്; അതായത് leading diagonal-ന് ഇരുവശവുമുള്ള entries പരസ്പരം mirror images ആണ്.

A determinant $|a_{ij}|$ is **symmetric** if $a_{ij} = a_{ji}$ for all i, j : the entries are mirror images across the main (leading) diagonal. Two properties: the square of a symmetric determinant is symmetric, and the adjoint of a symmetric determinant is symmetric.

മലയാളത്തിൽ

Main diagonal ഒരു കണ്ണാടി ആണെന്ന് കരുതുക; അതിന്റെ ഒരു വശത്തുള്ള numbers മറുവശത്ത് അതേപോലെ പ്രതിഫലിക്കുന്നതിൽ determinant symmetric ആണ്. അതായത് (1, 2)-ലെ number തന്നെ (2, 1)-ലും, (1, 3)-ലേത് (3, 1)-ലും. ഇങ്ങനെയുള്ളതിന്റെ square-ഉം adjoint-ഉം symmetric തന്നെ ആയിരിക്കും.

Symmetric: $a_{ij} = a_{ji}$ for all i, j

General form:
$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

(i) Δ^2 is symmetric. (ii) The adjoint of Δ is symmetric.

EXAMPLE (IGNOU EXAMPLE 16)

For $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 7 \end{vmatrix}$, find Δ^2 and $\text{adj } \Delta$.

$\Delta^2 = \Delta \times \Delta$ (row by column). Entry (1, 1) = $1 + 4 + 9 = 14$; (1, 2) = $2 + 8 + 15 = 25$; (1, 3) = $3 + 10 + 21 = 34$; (2, 2) = $4 + 16 + 25 = 45$; (2, 3) = $6 + 20 + 35 = 61$; (3, 3) = $9 + 25 + 49 = 83$.

$\Delta^2 = \begin{vmatrix} 14 & 25 & 34 \\ 25 & 45 & 61 \\ 34 & 61 & 83 \end{vmatrix}$, which is symmetric.

Cofactors: $A_1 = 28 - 25 = 3$, $B_1 = -(14 - 15) = 1$, $C_1 = 10 - 12 = -2$, $A_2 = -(14 - 15) = 1$, $B_2 = 7 - 9 = -2$, $C_2 = -(5 - 6) = 1$, $A_3 = 10 - 12 = -2$, $B_3 = -(5 - 6) = 1$, $C_3 = 4 - 4 = 0$.

$\text{adj } \Delta = \begin{vmatrix} 3 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 0 \end{vmatrix}$, which is symmetric.

Values: $\Delta = 1(3) + 2(1) + 3(-2) = -1$, so Δ^2 should equal 1. Indeed $14(45 \cdot 83 - 61 \cdot 61) - 25(25 \cdot 83 - 61 \cdot 34) + 34(25 \cdot 61 - 45 \cdot 34) = 14(14) - 25(1) + 34(-5) = 196 - 25 - 170 = 1$. $\checkmark$

Skew and skew-symmetric determinants

Skew determinant: A determinant with $a_{ij} = -a_{ji}$ for all $i \neq j$; the diagonal elements may be anything.

$i \neq j$ ആയ എല്ലാ i, j -നും $a_{ij} = -a_{ji}$ ആയ determinant; diagonal elements എന്തുമാകാം.

Skew-symmetric determinant: A skew determinant whose diagonal elements are all 0.

Diagonal elements എല്ലാം 0 ആയ skew determinant.

A determinant is **skew** if $a_{ij} = -a_{ji}$ for all $i \neq j$ (off-diagonal entries are negatives of their mirror images; the diagonal can be anything). It is **skew-symmetric** if, in addition, every diagonal entry is 0. A skew-symmetric determinant of odd order is always 0; one of even order is a perfect square.

മലയാളത്തിൽ

Skew-ൽ diagonal-ന് അപ്പുറവും ഇപ്പുറവും ഉള്ള numbers ഒരേ വലിപ്പമാണ്, പക്ഷേ sign മറിച്ച്. Diagonal കൂടി മുഴുവൻ zero ആണെങ്കിൽ അത് skew-symmetric. Odd order (3×3 പോലെ) skew-symmetric determinant എപ്പോഴും zero ആണ് – ഇത് ഓർത്തുവെച്ചാൽ exam-ൽ ഒറ്റ വരിയിൽ ഉത്തരം എഴുതാം. Even order ആണെങ്കിൽ ഉത്തരം ഒരു perfect square ആയിരിക്കും.

Skew: $a_{ij} = -a_{ji}$ ($i \neq j$), e.g.
$$\begin{vmatrix} a & -h & -g \\ h & b & -f \\ g & f & c \end{vmatrix}$$

Skew-symmetric: $a_{ij} = -a_{ji}$ and $a_{ii} = 0$, e.g.
$$\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$$

Odd order skew-symmetric $\Rightarrow$ value 0

Even order skew-symmetric $\Rightarrow$ value is a perfect square, e.g.
$$\begin{vmatrix} 0 & a \\ -a & 0 \end{vmatrix} = a^2$$

EXAMPLE (IGNOU EXAMPLE 17)

Show that $\Delta = \begin{vmatrix} 0 & 5 & 2 \\ -5 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = 0$.

Take -1 out of each of the three columns: $\Delta = (-1)^3 \begin{vmatrix} 0 & -5 & -2 \\ 5 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix}$.

The new determinant is the transpose of Δ , so its value is Δ (Property I). Hence $\Delta = -\Delta \Rightarrow 2\Delta = 0 \Rightarrow \Delta = 0$.

Direct check along R_1 : $0 - 5(0 - 6) + 2(-15 - 0) = 30 - 30 = 0$. $\checkmark$

EXAMPLE (EVEN ORDER)

$$\Delta_1 = \begin{vmatrix} 0 & 2 \\ -2 & 0 \end{vmatrix} = 0 - (-4) = 4 = 2^2, \text{ a perfect square.}$$

Exam tip: The same trick proves the general result: for odd n , taking -1 out of every column gives $\Delta = (-1)^n \Delta = -\Delta$, so $\Delta = 0$. This does not work for even n because $(-1)^n = 1$.

Cramer's rule for two equations

Definition: For $a_1x + b_1y = k_1$, $a_2x + b_2y = k_2$ with coefficient determinant $\Delta \neq 0$, $x = \Delta_1/\Delta$ and $y = \Delta_2/\Delta$, where Δ_1, Δ_2 are Δ with the x - and y -columns replaced by the constants.

$a_1x + b_1y = k_1$, $a_2x + b_2y = k_2$ എന്നതിൽ coefficient determinant $\Delta \neq 0$ ആണെങ്കിൽ $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$; ഇവിടെ Δ_1, Δ_2 എന്നിവ Δ -യിലെ x -column-ഉം y -column-ഉം constants കൊണ്ട് മാറ്റിയ determinants ആണ്.

For $a_1x + b_1y = k_1$, $a_2x + b_2y = k_2$, let Δ be the coefficient determinant. Δ_1 is Δ with the x -column replaced by the constants, and Δ_2 is Δ with the y -column replaced by the constants. If $\Delta \neq 0$ the system has the unique solution $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$. If $\Delta = 0$ the rule cannot divide, and you check Δ_1, Δ_2 .

മലയാളത്തിൽ

Cramer's rule-ൽ മൂന്ന് determinants കണ്ടെത്തണം: Δ (coefficients മാത്രം), Δ_1 (x-ന്റെ column-ന് പകരം constants), Δ_2 (y-ന്റെ column-ന് പകരം constants).

പിന്നെ $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$. $\Delta = 0$ ആണെങ്കിൽ ഹരിക്കാൻ പറ്റില്ല; അപ്പോൾ Δ_1, Δ_2 എന്തെങ്കിലും non-zero ആണെങ്കിൽ solution ഇല്ല, എല്ലാം zero ആണെങ്കിൽ (രണ്ട് variables-ൽ) infinitely many solutions.

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \Delta_1 = \begin{vmatrix} k_1 & b_1 \\ k_2 & b_2 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & k_1 \\ a_2 & k_2 \end{vmatrix}$$

$$\Delta \neq 0 \Rightarrow \text{unique solution } x = \Delta_1/\Delta, y = \Delta_2/\Delta$$

$$\Delta = 0 \text{ and some } \Delta_i \neq 0 \Rightarrow \text{no solution (inconsistent)}$$

$$\Delta = 0 \text{ and } \Delta_1 = \Delta_2 = 0 \Rightarrow \text{infinitely many solutions (for 2 equations, with not all coefficients 0)}$$

EXAMPLE (IGNOU EXAMPLE 18)

Solve $x_1 - 2x_2 = 3$, $3x_1 + 5x_2 = 20$.

$$\Delta = \begin{vmatrix} 1 & -2 \\ 3 & 5 \end{vmatrix} = 5 + 6 = 11 \neq 0.$$

$$\Delta_1 = \begin{vmatrix} 3 & -2 \\ 20 & 5 \end{vmatrix} = 15 + 40 = 55, \Delta_2 = \begin{vmatrix} 1 & 3 \\ 3 & 20 \end{vmatrix} = 20 - 9 = 11.$$

$$x_1 = 55/11 = 5, x_2 = 11/11 = 1. \text{ Check: } 5 - 2 = 3 \checkmark, 15 + 5 = 20 \checkmark.$$

EXAMPLE (CYP 3, Q1: ALL THREE CASES)

(i) $3x + 5y = -11$, $2x - 3y = 18$. $\Delta = \begin{vmatrix} 3 & 5 \\ 2 & -3 \end{vmatrix} = -9 - 10 = -19 \neq 0$. $\Delta_1 = \begin{vmatrix} -11 & 5 \\ 18 & -3 \end{vmatrix} = 33 - 90 = -57$, $\Delta_2 = \begin{vmatrix} 3 & -11 \\ 2 & 18 \end{vmatrix} = 54 + 22 = 76$. $x = -57/-19 = 3$, $y = 76/-19 = -4$. Check: $9 - 20 = -11 \checkmark$.

(ii) $x - 2y = 5$, $-2x + 4y = 8$. $\Delta = \begin{vmatrix} 1 & -2 \\ -2 & 4 \end{vmatrix} = 4 - 4 = 0$. $\Delta_1 = \begin{vmatrix} 5 & -2 \\ 8 & 4 \end{vmatrix} = 20 + 16 = 36 \neq 0 \Rightarrow \text{no solution.}$

(Doubling the first equation gives $-2x + 4y = -10$, which contradicts $= 8$.)

(iii) $2x - y = 6$, $-6x + 3y = -18$. $\Delta = \begin{vmatrix} 2 & -1 \\ -6 & 3 \end{vmatrix} = 6 - 6 = 0$, $\Delta_1 = \begin{vmatrix} 6 & -1 \\ -18 & 3 \end{vmatrix} = 18 - 18 = 0$, $\Delta_2 = \begin{vmatrix} 2 & 6 \\ -6 & -18 \end{vmatrix} = -36 + 36 = 0 \Rightarrow \text{infinitely many solutions. Put } y = k: 2x = 6 + k \Rightarrow x = (6 + k)/2, y = k, k \text{ any real number.}$

Cramer's rule for three equations

Definition: For three linear equations in x, y, z with coefficient determinant $\Delta \neq 0$, $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$, $z = \Delta_3/\Delta$, where $\Delta_1, \Delta_2, \Delta_3$ are Δ with the x, y, z columns replaced by the constants.

x, y, z -ൽ ഉള്ള മൂന്ന് linear equations-ന്റെ coefficient determinant $\Delta \neq 0$ ആണെങ്കിൽ $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$, $z = \Delta_3/\Delta$; ഇവിടെ $\Delta_1, \Delta_2, \Delta_3$ എന്നിവ Δ -യിലെ x, y, z columns constants കൊണ്ട് മാറ്റിയ determinants ആണ്.

For $a_i x + b_i y + c_i z = k_i$ ($i = 1, 2, 3$), replace the x, y, z columns of Δ in turn by the constants to get $\Delta_1, \Delta_2, \Delta_3$. The proof is short: $x\Delta$ equals Δ with the first column multiplied by x ; then $C_1 \rightarrow C_1 + yC_2 + zC_3$ turns that column into (k_1, k_2, k_3) , so $x\Delta = \Delta_1$.

മലയാളത്തിൽ

മൂന്ന് variables ആകുമ്പോൾ നാല് determinants: $\Delta, \Delta_1, \Delta_2, \Delta_3$. ഓരോന്നിലും ആ variable-ന്റെ column മാത്രം constants കൊണ്ട് മാറ്റുക, ബാക്കി columns അതേപടി. ഉത്തരം കിട്ടിയാൽ മൂന്ന് equations-ലും substitute ചെയ്ത് നോക്കുന്നത് ഒരു നല്ല ശീലമാണ് – ഒരു മിനിറ്റിൽ തെറ്റ് കണ്ടുപിടിക്കാം.

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \Delta_1 = \begin{vmatrix} k_1 & b_1 & c_1 \\ k_2 & b_2 & c_2 \\ k_3 & b_3 & c_3 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & k_1 & c_1 \\ a_2 & k_2 & c_2 \\ a_3 & k_3 & c_3 \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_1 & b_1 & k_1 \\ a_2 & b_2 & k_2 \\ a_3 & b_3 & k_3 \end{vmatrix}$$

$\Delta \neq 0 \Rightarrow x = \Delta_1/\Delta, y = \Delta_2/\Delta, z = \Delta_3/\Delta$ (unique solution)

$\Delta = 0$, some $\Delta_i \neq 0 \Rightarrow$ no solution

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0 \Rightarrow$ no unique solution: find it by setting one variable = k and solving two equations (then check the third)

EXAMPLE (IGNOU EXAMPLE 19)

Solve $2x + 3y + z = 4$, $x - y + 2z = 9$, $3x + 2y - z = 1$.

$$\Delta = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{vmatrix} = 2(1 - 4) - 3(-1 - 6) + 1(2 + 3) = -6 + 21 + 5 = 20 \neq 0, \text{ so the solution is unique.}$$

$$\Delta_1 = \begin{vmatrix} 4 & 3 & 1 \\ 9 & -1 & 2 \\ 1 & 2 & -1 \end{vmatrix} = 4(1 - 4) - 3(-9 - 2) + 1(18 + 1) = -12 + 33 + 19 = 40.$$

$$\Delta_2 = \begin{vmatrix} 2 & 4 & 1 \\ 1 & 9 & 2 \\ 3 & 1 & -1 \end{vmatrix} = 2(-9 - 2) - 4(-1 - 6) + 1(1 - 27) = -22 + 28 - 26 = -20.$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 4 \\ 1 & -1 & 9 \\ 3 & 2 & 1 \end{vmatrix} = 2(-1 - 18) - 3(1 - 27) + 4(2 + 3) = -38 + 78 + 20 = 60.$$

$$x = 40/20 = 2, y = -20/20 = -1, z = 60/20 = 3.$$

Check: $4 - 3 + 3 = 4 \checkmark$; $2 + 1 + 6 = 9 \checkmark$; $6 - 2 - 3 = 1 \checkmark$.

EXAMPLE (IGNOU EXAMPLE 20: INFINITELY MANY SOLUTIONS)

Solve $x + 3y + 2z = 6$, $-x + 4y + 5z = 8$, $2x + 5y + 3z = 10$.

$$\Delta = \begin{vmatrix} 1 & 3 & 2 \\ -1 & 4 & 5 \\ 2 & 5 & 3 \end{vmatrix} = 1(12 - 25) - 3(-3 - 10) + 2(-5 - 8) = -13 + 39 - 26 = 0.$$

$$\Delta_1 = \begin{vmatrix} 6 & 3 & 2 \\ 8 & 4 & 5 \\ 10 & 5 & 3 \end{vmatrix} = 6(12 - 25) - 3(24 - 50) + 2(40 - 40) = -78 + 78 + 0 = 0.$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 2 \\ -1 & 8 & 5 \\ 2 & 10 & 3 \end{vmatrix} = 1(24 - 50) - 6(-3 - 10) + 2(-10 - 16) = -26 + 78 - 52 = 0.$$

$$\Delta_3 = \begin{vmatrix} 1 & 3 & 6 \\ -1 & 4 & 8 \\ 2 & 5 & 10 \end{vmatrix} = 1(40 - 40) - 3(-10 - 16) + 6(-5 - 8) = 0 + 78 - 78 = 0.$$

All are 0. Put $z = k$ in the first two equations: $x + 3y = 6 - 2k$, $-x + 4y = 8 - 5k$.

$$\text{Now } \Delta = \begin{vmatrix} 1 & 3 \\ -1 & 4 \end{vmatrix} = 4 + 3 = 7, \Delta_1 = \begin{vmatrix} 6 - 2k & 3 \\ 8 - 5k & 4 \end{vmatrix} = 24 - 8k - 24 + 15k = 7k, \Delta_2 = \begin{vmatrix} 1 & 6 - 2k \\ -1 & 8 - 5k \end{vmatrix} = 8 - 5k + 6 - 2k = 14 - 7k.$$

$$x = 7k/7 = k, y = (14 - 7k)/7 = 2 - k, z = k.$$

Check in the third equation: $2k + 5(2 - k) + 3k = 10 \checkmark$. So $x = k$, $y = 2 - k$, $z = k$ for every real k .

EXAMPLE (CYP 3, Q2: NO SOLUTION)

Solve $x + y + 2z = 4$, $x - y + 3z = 3$, $2x + 2y + 4z = 7$.

$$\Delta = \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 2 & 4 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 1 & 2 \end{vmatrix} = 0 \text{ (take 2 from } R_3; \text{ then } R_1 = R_3).$$

$$\Delta_1 = \begin{vmatrix} 4 & 1 & 2 \\ 3 & -1 & 3 \\ 7 & 2 & 4 \end{vmatrix} = 4(-4 - 6) - 1(12 - 21) + 2(6 + 7) = -40 + 9 + 26 = -5 \neq 0.$$

$\Delta = 0$ but $\Delta_1 \neq 0$, so there is no solution. (Indeed, doubling equation 1 gives $2x + 2y + 4z = 8$, not 7.)

EXAMPLE (CYP 3, Q3)

Solve $x + 2y + 3z = 6$, $2x + 4y + z = 7$, $3x + 2y + 9z = 14$.

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 2 & 9 \end{vmatrix} = 1(36 - 2) - 2(18 - 3) + 3(4 - 12) = 34 - 30 - 24 = -20 \neq 0.$$

$$\Delta_1 = \begin{vmatrix} 6 & 2 & 3 \\ 7 & 4 & 1 \\ 14 & 2 & 9 \end{vmatrix} = 6(36 - 2) - 2(63 - 14) + 3(14 - 56) = 204 - 98 - 126 = -20.$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 3 \\ 2 & 7 & 1 \\ 3 & 14 & 9 \end{vmatrix} = 1(63 - 14) - 6(18 - 3) + 3(28 - 21) = 49 - 90 + 21 = -20.$$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 3 & 2 & 14 \end{vmatrix} = 1(56 - 14) - 2(28 - 21) + 6(4 - 12) = 42 - 14 - 48 = -20.$$

$x = y = z = -20/-20 = 1$. Check: $1 + 2 + 3 = 6 \checkmark$, $2 + 4 + 1 = 7 \checkmark$, $3 + 2 + 9 = 14 \checkmark$.

Exam tip: For three equations, $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ does not by itself guarantee infinitely many solutions (e.g. $x + y + z = 1$, $x + y + z = 2$, $x + y + z = 3$ has all four determinants 0 but no solution). So always check your parametric answer in the equation you did not use, as done in Example 20. In the printed CYP 3, Q3 the "2y" terms appear as "24" (an extraction error); the system is $x + 2y + 3z = 6$, $2x + 4y + z = 7$, $3x + 2y + 9z = 14$.

Quick recap

- A determinant is a scalar defined only for a square matrix; order n has $n!$ terms.
- $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$; for 1×1 , $|[a]| = a$ (not the modulus).
- Minor M_{ij} : delete row i and column j . Cofactor $C_{ij} = (-1)^{i+j}M_{ij}$.
- $|A|$ = sum of (element $\times$ cofactor) along any one row or column; pick the line with most zeros.
- Transpose: value unchanged. Swap two lines: sign changes.
- Two identical or proportional lines, or a zero line $\Rightarrow$ value 0.
- A common factor of one line comes outside; $|kA| = k^n|A|$ for an $n \times n$ matrix.
- $R_i \rightarrow R_i + kR_j$ does not change the value: the main tool for making zeros.
- A line whose entries are sums splits the determinant into a sum of two determinants.
- Same-order determinants multiply like matrices; $|AB| = |A||B|$.
- Adjoint: replace each element by its cofactor; its value is Δ^2 (order 3). Reciprocal = adjoint $\div \Delta$, value $1/\Delta$.

- $\text{adj } A = \text{transpose of the cofactor matrix}$; $A^{-1} = \text{adj } A / |A|$, only if $|A| \neq 0$.
- Symmetric: $a_{ij} = a_{ji}$; its square and adjoint are symmetric. Skew-symmetric ($a_{ij} = -a_{ji}$, zero diagonal): odd order $\Rightarrow 0$, even order $\Rightarrow$ perfect square.
- Cramer's rule: $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$, $z = \Delta_3/\Delta$ when $\Delta \neq 0$; $\Delta = 0$ with some $\Delta_i \neq 0 \Rightarrow$ no solution.

Check yourself

Q. 1. Evaluate $\begin{vmatrix} 4 & -3 \\ 2 & 5 \end{vmatrix}$ and find x if $\begin{vmatrix} x & 4 \\ 4 & x \end{vmatrix} = 0$.

$$4 \times 5 - 2 \times (-3) = 20 + 6 = 26.$$

$$x^2 - 16 = 0 \Rightarrow x = 4 \text{ or } x = -4.$$

Q. 2. For $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$, find $M_{21}, C_{21}, M_{32}, C_{32}$.

$$M_{21} = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2; C_{21} = (-1)^3(-2) = 2.$$

$$M_{32} = \begin{vmatrix} 1 & 1 \\ 2 & -3 \end{vmatrix} = -3 - 2 = -5; C_{32} = (-1)^5(-5) = 5.$$

Q. 3. Evaluate $\begin{vmatrix} 2 & 0 & 5 \\ 1 & 0 & 3 \\ 4 & 7 & -1 \end{vmatrix}$, choosing the best line to expand.

C_2 has two zeros. Only the element 7 at position (3, 2) contributes, with sign $(-1)^{3+2} = -1$.

$$\text{Value} = -7 \begin{vmatrix} 2 & 5 \\ 1 & 3 \end{vmatrix} = -7(6 - 5) = -7.$$

Q. 4. Without expanding, show $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$.

$C_3 \rightarrow C_3 + C_2$: every entry of C_3 becomes $a + b + c$.

Take $(a + b + c)$ out: C_3 becomes $(1, 1, 1) = C_1$, so the value is $(a + b + c) \times 0 = 0$.

Q. 5. Prove using properties:
$$\begin{vmatrix} a+b & c & c \\ a & b+c & a \\ b & b & c+a \end{vmatrix} = 4abc \text{ (CYP 2, Q3).}$$

$R_1 \rightarrow R_1 - R_2 - R_3: R_1 = (0, -2b, -2a).$

$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1: \begin{vmatrix} 0 & -2b & -2a \\ a & b+c-a & 0 \\ b & 0 & c+a-b \end{vmatrix}.$

Expand along $R_1: 0 - (-2b)[a(c+a-b) - 0] + (-2a)[0 - b(b+c-a)] = 2ab(c+a-b) + 2ab(b+c-a) = 2ab(2c) = 4abc.$

Check with $a = 1, b = 2, c = 3$: direct value $24 = 4 \times 6. \checkmark$

Q. 6. Prove:
$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1+x^3 & 1+y^3 & 1+z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(1+xyz) \text{ (CYP 2, Q2).}$$

Split R_3 (Property V): $\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix}.$

In the second, take x, y, z out of $C_1, C_2, C_3: xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}.$

In the first, $R_1 \leftrightarrow R_3$ then $R_2 \leftrightarrow R_3$ (two swaps, sign $(-1)(-1) = +1$) gives the same determinant $D =$

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}.$$

So $\Delta = (1+xyz)D$. D is the transpose of the Vandermonde determinant, so $D = (x-y)(y-z)(z-x)$. Hence $\Delta = (x-y)(y-z)(z-x)(1+xyz)$.

Q. 7. Find the adjoint and inverse of $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$.

$$|A| = 8 - 3 = 5 \neq 0.$$

Cofactors: $C_{11} = 4$, $C_{12} = -1$, $C_{21} = -3$, $C_{22} = 2$. Cofactor matrix $\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$, so $\text{adj } A = \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$ (swap the diagonal, negate the others).

$$A^{-1} = (1/5) \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}. \text{ Check: } A \cdot \text{adj } A = \begin{pmatrix} 8 - 3 & -6 + 6 \\ 4 - 4 & -3 + 8 \end{pmatrix} = 5I. \checkmark$$

Q. 8. Without expanding, find $\begin{vmatrix} 0 & p & -q \\ -p & 0 & r \\ q & -r & 0 \end{vmatrix}$.

$a_{ij} = -a_{ji}$ and all diagonal entries are 0, so it is skew-symmetric of odd order (3). Its value is 0.

$$\text{Check: } 0 - p(0 - qr) + (-q)(pr - 0) = pqr - pqr = 0. \checkmark$$

Q. 9. Solve by Cramer's rule: $2x + y = 5$, $x - y = 1$.

$$\Delta = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -2 - 1 = -3 \neq 0.$$

$$\Delta_1 = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -5 - 1 = -6, \Delta_2 = \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = 2 - 5 = -3.$$

$$x = -6/-3 = 2, y = -3/-3 = 1. \text{ Check: } 4 + 1 = 5 \checkmark, 2 - 1 = 1 \checkmark.$$

Q. 10. Solve by Cramer's rule: $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$.

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 1(1 - 1) - 1(-1 - 2) + 1(1 + 2) = 0 + 3 + 3 = 6.$$

$$\Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 6(1 - 1) - 1(-2 - 1) + 1(2 + 1) = 0 + 3 + 3 = 6.$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 1(-2 - 1) - 6(-1 - 2) + 1(1 - 4) = -3 + 18 - 3 = 12.$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix} = 1(-1 - 2) - 1(1 - 4) + 6(1 + 2) = -3 + 3 + 18 = 18.$$

$x = 6/6 = 1$, $y = 12/6 = 2$, $z = 18/6 = 3$. Check: $1 + 2 + 3 = 6 \checkmark$, $1 - 2 + 3 = 2 \checkmark$, $2 + 2 - 3 = 1 \checkmark$.

Based on the IGNOU unit texts. Check definitions against the PDFs before the exam.